



**BERLIN SCHOOL OF
BUSINESS & INNOVATION**

**Dissertation Title: Optimizing Multi-Cloud Environments Through
DevOps: A Framework for SMEs**

Master title: Information Technology Management

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ABSTRACT

This work is aimed to explore the optimal multi-cloud environment adopting DevOps practises into SMEs. As SMEs have more and more reliance on cloud infrastructure, it presents significant challenges for them in terms of cost, performance and security. We performed a quantitative survey with 100 IT and cloud practitioners to better understand the impact of DevOps on performance, scalability, and cost efficiency in multi-cloud environments. The average level of DevOps adoption was moderate based on descriptive statistics, combined with high use of the biggest cloud providers e.g. AWS and Azure. Correlation and regression between DevOps practice and cloud performance revealed a weak statistical association, implying that the benefits may not be exclusively as due to the technical implementation but become effective only in relation to organization's maturity. ANOVA test results further indicated no significant variation in performance over the cloud providers. While there's little to no statistical impact, DevOps is still a strategic play for automation, collaboration and operational efficiency. The paper provides pragmatic observations and suggestions for SMEs to design efficient multi-cloud plans.

Keywords: DevOps Practices, Multi-Cloud Adoption, SMEs (Small and Medium Enterprises), Cloud Performance Optimization, Cost Efficiency

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Gaddala Lokesh Kumar

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DISSERTATION THESIS



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INTRODUCTION

Brief Research of the Topic

Flexibility, scalability, and cost-efficiency in multi-cloud settings are necessitating a quick adoption of this type of environment, especially by Small and Medium-sized Enterprises (SMEs) (Merseedi and Zeebaree, 2024). Multi-cloud strategies imply the employment of services of many cloud providers to allow businesses to prevent vendor lock-in and make the most of each cloud provider services (Chen et al., 2024). The problem, however, is that it is difficult to manage these resources efficiently, particularly on behalf of SMEs having limited expertise and resources. It is possible to incorporate the DevOps practices into the multi-cloud environments to simplify work, automatize the processes, and increase security and, therefore, the overall efficiency (Sandhu et al., 2024). In addition to these strengths, SMEs are challenged by such obstacles as high start-up expenses and shortage of skilled labor (Hemalatha et al., 2024).

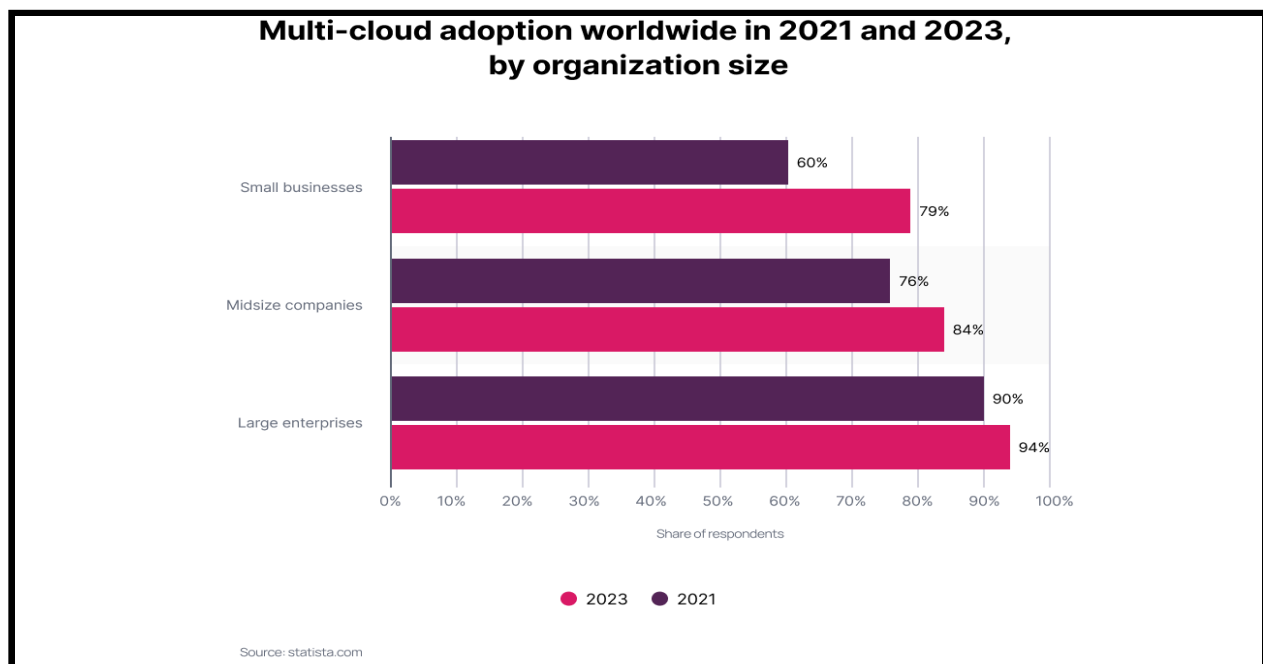


Figure 1: Multi-cloud Adoption Worldwide in 2021 and 2023, by Organization Size

(Source: <https://www.statista.com/statistics/1245569/multi-cloud-adoption-organization-size-worldwide>)

The rise in multi-cloud adoption has been pointed out in 2021-2023 as noted in Figure 1. The shares of small businesses increased by 60-79%, medium-sized companies increased by 76-84%,

and big businesses increased by 90-94%. This growth highlights why optimized frameworks (particularly in the case of SMEs) are necessary to successfully apply the multi-cloud strategies.

Triggers and Rationale

The growing adoption of cloud computing by SMEs has raised the issue of effective management of multi-cloud environments, particularly given the growing complexity and resource demands (Merseedi and Zeebaree, 2024). One of the possible solutions to the problem of maximizing the use of cloud resources, automatizing operations, and improving their efficiency is implementing the DevOps practices as part of the multi-cloud strategies (Chen et al., 2024). The justification of this study has to do with the fact that SMEs tend to grapple with the process of integrating DevOps within a multi-cloud environment and therefore inefficiencies and high costs ensue (Hemalatha et al., 2024). The paper under discussion seeks to examine how SMEs can streamline their multi-clouds with the help of DevOps, which will take part in a more efficient and cost-effective and secure way of handling clouds (Gudelli, 2024).

Research Gaps

Although the literature shows the benefits of multi-cloud and DevOps in large companies, the gap in current research on this area is evident in the information available on SMEs, especially the opportunities and challenges that these companies have when implementing such technologies (Merseedi and Zeebaree, 2024). Also, few researches investigate the specifics of the structure to optimize multi-cloud environment using DevOps in the SME environment (Sandhu et al., 2024). Empirical data on the ways SMEs may overcome the need to balance performance, cost efficiency, and security whilst expanding their cloud infrastructures is also lacking (Chen et al., 2024). The proposed research seeks to fill these gaps as well as provide practicable solutions to SMEs.

Aims and Objectives

The study will attempt to understand the role of optimization of SMEs multi-cloud environments by integrating DevOps practices, which will help to achieve maximum cost efficiency, performance, and security and overcome main implementation issues. The objectives are:

- RO1: To examine the effect of DevOps on the cost effectiveness and performance of multi-cloud within SMEs.
- RO2: To determine the major issues SMEs encounter in their quest to adopt multi-cloud strategies and DevOps approaches.

- RO3: To evaluate the potential of cloud resource optimization techniques, using the concept of DevOps, in improving the scalability and security in multi-cloud architectures in the context of SMEs.

Research Questions

RQ1: How do DevOps practices influence the performance and cost efficiency of multi-cloud environments in small and medium-sized enterprises (SMEs)?

RQ2: What are the key challenges SMEs face when adopting multi-cloud environments and implementing DevOps practices in their cloud infrastructure?

RQ3: To what extent can cloud resource optimization strategies, based on DevOps principles, improve the scalability and security of multi-cloud infrastructures in SMEs?

Methodology

The proposed study will be quantitative and primary with the sample of 100 SMEs, which are in particular, key decision-makers, including IT managers, cloud architects, and DevOps practitioners, in these organizations. These people will be chosen according to their participation in the realization and organization of multi-cloud system and DevOps practices. The statistical methods will be used to analyze the data with the intention of identifying patterns and correlations. Regression analysis, correlation tests, and descriptive statistics will be performed with the help of such tools as stats kingdom and excel, which will allow comprehending better performance, cost efficiency, scalability, and security issues in multi-cloud settings.

Structure of the thesis

Chapter 1: Literature Review 1

This part is a brief introduction to multi-cloud environment, and its increased relevance in SMEs. It discusses the fundamental aspects of multi-cloud strategies and its advantages, such as cost optimization, flexibility, and reduction of risks. Also, this chapter includes the role of cloud service providers and how SMEs can use these strategies to satisfy their individual needs.

Chapter 2: Literature Review 2

The second chapter explores the topic of integrating the DevOps practice in the multi-cloud environments in greater detail. It examines the available structures and practices that integrate DevOps and multi-cloud approaches centered on performance, scalability, and security in SMEs.

Chapter 3: Methodology

It provides the research methodology, which is based on secondary research by analyzing data on Kaggle datasets. It describes how the data will be collected, selection criteria, as well as how literature will be reviewed.

Chapter 4: Findings, Analysis, and Discussion

This part outlines the results of the secondary data analysis, and the results are then discussed in details. It analyzes the information with regard to the research questions, bringing out trends, challenges, and opportunities on how to maximize multi-cloud environments with DevOps.

Conclusion: This section provides an overview of the main results of the study and their implication to the SMEs using multi-cloud environments with DevOps. It gives recommendations that SMEs need to streamline their cloud infrastructure and outlines research areas in the future.

Finally, this chapter examines how multi-cloud environments can be optimised at the level of SMEs by using DevOps practices to improve the cost-efficiency, performance, and security. Chapter 1 goes further to analyze the multi-cloud strategies and the applicability to the SMEs.

CHAPTER ONE – LITERATURE REVIEW I

1.1 Chapter Introduction

This literature review one explains why the concept of multi-clouds is increasingly becoming relevant to small and medium-sized enterprises (SMEs). Multi-clouds strategies which entail services of more than one cloud provider provide greater flexibility, scalability and cost-effectiveness to SMEs. These environments enable the businesses to escape the vendor lock-in and be able to optimize the cloud resources according to the requirements. Nonetheless, management of multi-cloud infrastructure presents a big challenge to the SMEs, especially on resource control, security and expertise constraints. This review will discuss the advantages, issues and role of cloud providers in multi-cloud adoption, thus providing the grounds on how these environments can be optimized using practical strategies which are specific to SMEs.

1.2 Introduction to Multi-Cloud Environments

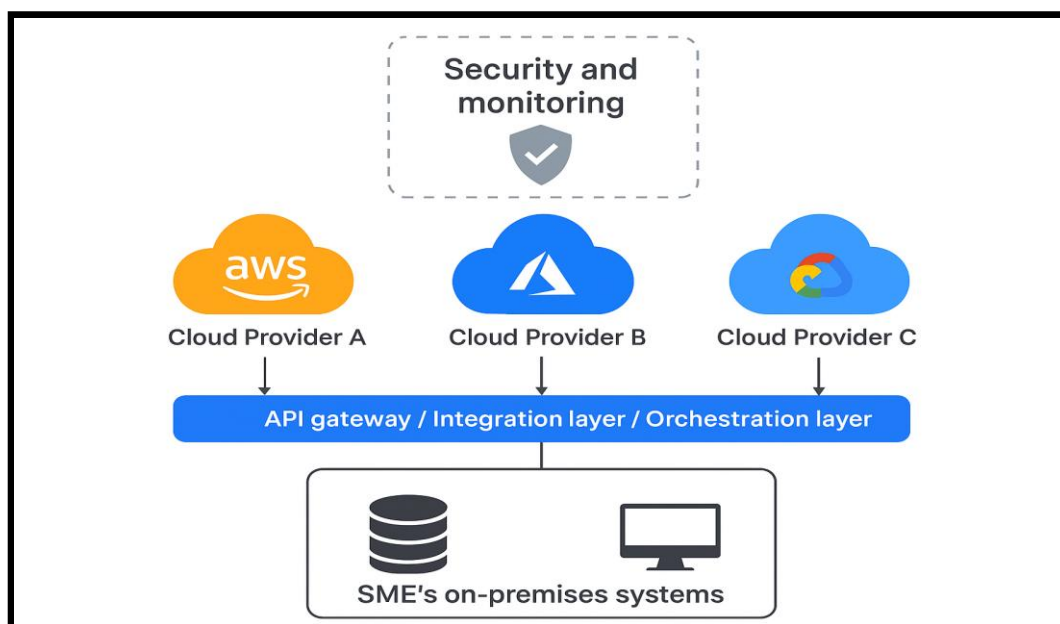


Figure 2: Multi-Cloud Environment Architecture

(Source: Created through Canva)

The following diagram (figure 2) shows the integration of various cloud providers, AWS, Azure, and Google Cloud using API gateway and orchestration layer. Multi-cloud environment is a concept used to implement cloud services of more than one cloud provider, which enables business to eliminate the phenomenon of vendor lock-in and optimal utilization of the resources

across platforms (Chen et al., 2024). With the help of a combination of the public, private, or hybrid cloud, the companies can customize their cloud infrastructure to meet the needs of particular performance, cost-efficiency, and compliance requirements (Merseedi and Zeebaree, 2024). Although multi-cloud strategies are flexible, they create complexities in the management of dissimilar services, integration and security. The increased reliance on the services of various cloud vendors needs more effective management systems, which SMEs frequently have not yet because of resource and expertise constraints. Cloud Management Plays like the Morpheus and Scalr assist in the harmonization of the management of various clouds, easing the efforts of SMEs. (Haddara et al., 2021). Therefore, a multi-cloud strategy definition is the key to optimized performance and the reduction of risks. Cloud Management Systems like Morpheus and Scalr enable SMEs to easily manage multi-cloud resources, have a single dashboard to control costs, security and performance of cloud providers.

1.2.1 Evolution and Adoption of Multi-Cloud Strategies

This transformation of the multi-cloud strategies and adoption has turned out to be a major change in the way businesses handle their cloud infrastructure. First, the ease of using one cloud service provider was preferred due to the cost-efficiency and convenience to manage. Nevertheless, at the moment when businesses became bigger, and their requirements were more complicated, the weaknesses of single-provider models were realized, and multi-cloud strategies emerged (Chen et al., 2024). The shift towards using many providers enables the business to prevent vendor lock-in, maximize their cloud resources, and make them more flexible. Merseedi and Zeebaree (2024) found that this tendency is predetermined by the necessity to be more operationally resilient and be capable of adjusting cloud environments to the requirements. Hybrid and multi-clouds deployments have been particularly noticeable over the last ten years as businesses are trying to avoid risk and business continuity (Granlund et al., 2021). Nevertheless, there are significant obstacles to SMEs employing multi-cloud strategies, and the main ones are the complexity of managing multiple cloud systems and the technical skills needed to facilitate a smooth integration (Sandhu et al., 2024). The shortage of qualified specialists and difficulties in harmonizing the services of various providers intimidate the SMEs in the context of the actual implementation of the multi-cloud environment. Thus, the positive aspects of the adoption of multi-cloud are obvious, but the complications in its functioning cannot be neglected. To

leverage the benefits of multi-cloud, SMEs could combine tools such as Jenkins to facilitate CI/CD, Kubernetes to facilitate container orchestration, and Terraform to facilitate Infrastructure as Code (IaC). These can simplify the operations in the cloud, automate and make it scalable.

1.2.2 Importance of Multi-Cloud in the Modern Business Landscape

Multi-cloud strategies play the critical role to improve the operational resilience, flexibility, and scaling of businesses in the increasingly competitive world (Haddara et al., 2021). With the help of services provided by several cloud providers, businesses are able to optimize the usage of their resources, making sure that they are not restricted to one vendor, thereby cutting the risk of vendor lock-in (Merseedi and Zeebaree, 2024). In the case of SMEs, multi-cloud strategies are especially important because there is an availability of a wider variety of technologies and services, which would help them remain competitive by enhancing the quality of customer service and aligning their IT infrastructure with a particular need (Chen et al., 2024). Nevertheless, in spite of obvious advantages of using multi-clouds, there are also real challenges, especially on managerial and security levels. To deal with the security threats, SMEs may implement DevSecOps tools like Snyk and Aqua Security that provide vulnerabilities to the CI/CD pipeline and, therefore, vulnerabilities are identified early in the development lifecycle. The overall management of any type of cloud services may be complicated, and it may need strong integration and orchestration, which SMEs simply do not have the resources to manage properly (Sandhu et al., 2024). The security of multi-cloud environments may be jeopardized by the fact that security provisions differ between different platforms. One of the ways of these risks can be reduced by using DevSecOps tools such as Snyk or Aqua Security that allow SMEs to incorporate security into the DevOps pipeline. The solution to these problems requires SMEs to have powerful structures, including DevOps, to automate operations, simplify cloud-hosting and improve the level of security (Sandhu et al., 2024). The practical implication is that although multi-cloud strategies are able to promote growth and innovation among SMEs, they need to invest into training, automation tools as well as security measures to ensure that they are able to reap maximum benefits of such environments. In its absence the costs and risks can supersede the benefits.

1.3 Relevance of Multi-Cloud for SMEs

Multi-cloud strategies are relevant to SMEs that have become pivots in making SMEs more resilient in terms of operations, flexible and scalable. With the ever-increasing pace of the digital transformation, SMEs are finding it important to exclude the existence of vendor lock-in and whether their infrastructure is capable of addressing the various needs of the business (Waseem et al., 2025). The adoption of multi-clouds also allows SMEs to choose the best solutions among the various vendors and hence maximize their cost and performance, and this is one of the factors that are of great importance to the resource-limited businesses (Ware, 2025). Moreover, multi-cloud systems enable the SMEs to share workloads among various platforms and minimize the threat of service failures and enhance overall business resiliency (L.C.P. et al., 2025). Although these benefits exist, SMEs have to overcome numerous challenges, among which are the technical challenges, scarcity of resources, and absence of specialized staff that can handle such variety of setting. Thus, the advantages of adoption of multi-clouds are significant, but their practical implementation is quite a challenge of SMEs that do not have the required capabilities and infrastructure.

1.3.1 Benefits and Challenges of Multi-Cloud for SMEs

Multi-cloud policies are very essential in offering SMEs with critical opportunities particularly on flexibility and risk management. Without dependence on one vendor, the SMEs will be able to choose cloud services that will suit their needs, which allows them to optimize their performance and cost-efficiency (Waseem et al., 2025). Scalability is also improved through this method and a business is able to scale up and down its resources very fast depending on the demand. In addition, multi-cloud strategies will enhance business continuity because they will spread the workloads among cloud providers and minimize the effects of possible disruptions (Ware, 2025). Nevertheless, such advantages are offset by major difficulties. Among the most noticeable are the difficulties of being able to manage several cloud environments, which involve advanced integration and orchestration (L.C.P. et al., 2025). The SMEs tend to experience challenges with efficient monitoring and data protection of the information across various platforms, which contributes to higher operational expenses and the lack of security. Also, the deficiency of expertise and competencies in resources makes the effective implementation and control of multi-cloud infrastructures even harder, which is a rather expensive and resource-consuming model among most of the SMEs (Waseem et al., 2025).

1.4 Flexibility, Scalability, and Cost Optimization for SMEs

Multi-cloud environments can provide a significant team of flexibility, scalability, and cost-efficiency to the SMEs, but the benefits of the solutions might be dwarfed by the difficulties they present in operations. Multi-cloud strategies allow companies to choose the services offered by other vendors to achieve the maximum performance and cost-efficiency in their case (Taqi et al., 2024). Flexibility is determined by the fact that it is possible to select the best cloud service based on the needs of each business activity, thereby increasing the flexibility that the organization has to deal with the market dynamism. Multi-cloud systems can bring SMEs to scale more efficiently in relation to demand, and are able to grow without excessive capital spending on initial scale (Sandhu et al., 2024). In terms of cost, multi-cloud solutions allow SMEs to maximize the cost of the clouds by choosing the best prices to various services (Razdar et al., 2025). Nevertheless, cost savings are achievable, however, in managing the complexity of multiple providers, SMEs have problems on managing the complexities consequently subjecting to additional administrative overhead as well as security risks. Hence, it is necessary to balance the management and control in order to achieve the desired benefits. In the case of SMEs, Jenkins is a DevOps tool that can be used to automate the build and deployment processes to achieve rapid release cycles. Kubernetes has the potential to enable to orchestrate containers across various cloud providers, whereas Terraform can enable to provide cloud infrastructure automatically.

1.5 Cloud Service Providers and Multi-Cloud Integration

It is even more complicated because major cloud service providers such as AWS, Microsoft Azure, and Google Cloud have their distinct advantages in multi-cloud settings and have certain weaknesses. Having such a wide range of services, AWS offers scalability and sophisticated analytics, so it is a reliable solution to companies that need international infrastructure and quick distribution of resources (Cheerla, 2024). Nevertheless, its payment schemes and the intricacy of its services may be too overwhelming to the SMEs, so cost optimization becomes a pressing issue. Conversely, Microsoft Azure participates in the extensive integration of Microsoft products to provide a native experience of a hybrid cloud to those businesses that have a presence within that ecosystem (Granlund et al., 2021). This integration makes it easier to

manage however it may restrict flexibility to organizations who want to utilize other non-Microsoft solutions. The high commitment to AI and machine learning distinguishes Google Cloud as a product with impressive data-driven application capacities, which is why it is a valuable option among the SMEs that prioritize innovation and big data (Chen et al., 2024). Nevertheless, even with those personal advantages, the multi-cloud strategies require a complex degree of coordination to seek a smooth integration. Organisations will have to be sensitive to security issues along with compatibility, and inexperience may hamper the successful use of these platforms (Cheenepalli et al., 2025). Hence, as these providers provide highly effective tools, SMEs need to consider their technical capabilities and approach to effective exploitation of multi-cloud environments.

1.6 Key Considerations for SME Cloud Strategies

One of the major considerations that SMEs cloud strategies should make is the complexity of the nature of the multi-cloud environment and how it can be aligned using the meager resources available in the SME. Using the Resource-Based View (RBV) theory, we can see that SMEs must employ the internal resource that they have to fully maximize their cloud strategy namely their understanding of business processes, position in the market and other competitive advantages. The problem is, however, that SMEs do not always estimate the level of sophistication needed to make an ideal allocation of cloud resources. Although the RBV focuses on the optimization of resources, it is often problematic here that SMEs can match the resources of the cloud with their strategic goals because they do not have the necessary technical knowledge internally (Cheerla, 2024). Furthermore, the theory of transaction cost argues that the cost aspect of dealing with multiple cloud providers such as integration and coordination costs may exceed the benefits expected unless a well-crafted plan is put in place. Practically, SMEs are likely to have difficulties reducing transaction costs because they do not have formalised governance systems, which create inefficiencies. Lacking significant in-house capabilities to combine and operate various cloud platforms, SMEs risk only having a small payoff on the investments in the cloud and being unable to achieve a complete potential of their multi-cloud strategies (Ali et al., 2025).

1.6.1 Importance of Customization and Flexibility

Multi-cloud strategies are also associated with flexibility and customization as important sources of value to SMEs. The contingency theory argues that organizations should adjust their plans to the peculiarities of the environment. As a SME, this implies that they have the flexibility to choose the optimum service of various providers due to the multi-cloud architectures (Gianassi et al., 2025), and thus, be compatible with diverse and changing business demands. Nonetheless, although flexibility is a desirable quality, building internal capacity that would allow dealing with the complexity of working with multiple cloud platforms is also a necessity. The theory of dynamic capabilities puts stress on the fact that the flexibility is not merely the question of choosing the best services but rather the capability of the reorganization of the cloud resources in accordance with the changes in business needs. This can be a big problem to the SMEs, whereby most of them do not have the technical know-how and the infrastructure to operate in such complicated environments (Gianassi et al., 2025). The intricacy of fitting multiple cloud services that have their own interface, security measures and data schemas usually leads to inefficiency and performance bottlenecks.

Additionally, the trade-off between customization and integration can be an impediment to the advantages of multi-cloud flexibility (Guo et al., 2025). SMEs may want to have a specific solution due to a particular operational requirement, however, it is difficult to achieve smooth interoperability between various cloud services. Such an inconsistency between the strategic goals and the operational capacities frequently causes the existing gap where the required amount of flexibility is undermined by the real-life challenges of operating various cloud platforms. Therefore, SMEs have to walk the fine line between the selection of tailored solutions and managing the issue of integrating and maintaining them.

1.6.2 Best Practices for Successful Multi-Cloud Adoption

An overview of the best practices in the adoption of multi-cloud has indicated that adoption of multi-cloud does not only need to adopt the technology but also a high level of organizational preparedness and flexibility. Technology-Organization-Environment (TOE) framework has seen the times change to support new dynamics of adopting the cloud. Recent research indicates that balancing these three factors, i.e. technology, organizational capacity and environmental factors, is essential to the SMEs. Management of multi-clouds and automation and orchestration tools,

such as, are essential to industrially manage multi-cloud environments but demand advanced technical skill and expensive training of personnel (Granlund et al., 2021). Lack of such an investment can result in inefficiencies in operations of the SMEs. Diffusion of Innovations theory is also quite applicable since it emphasizes that the perceived benefits, e.g., cost efficiency, and scalability, need to correspond to the established systems of the SMEs to be successful. Security and conformity, especially amongst multi-cloud systems, is persistent (Ali et al., 2025). Unless DevSecOps concepts are included at the very beginning, SMEs may fall prey to security risks that may compromise their multi-cloud strategy (Cheenepalli et al., 2025). Finally, to be adopted successfully depends on possibly integrating technology and organizational capacity and taking a proactive step with regard to security. The best practices focus on ongoing learning, integration of security, and strategic management to overcome the complexity of the multi-cloud environment and achieve long-term success of SMEs (Guo et al., 2025).

This figure 3 depicts the DevSecOps incorporates development, security, and operations in a continuous integration and deployment model. The outermost layer was associated with cloud providers and automation tools, which demonstrates the constant feedback loop necessary to achieve the security and optimization of SME multi-cloud systems by means of proactive collaboration and automation.

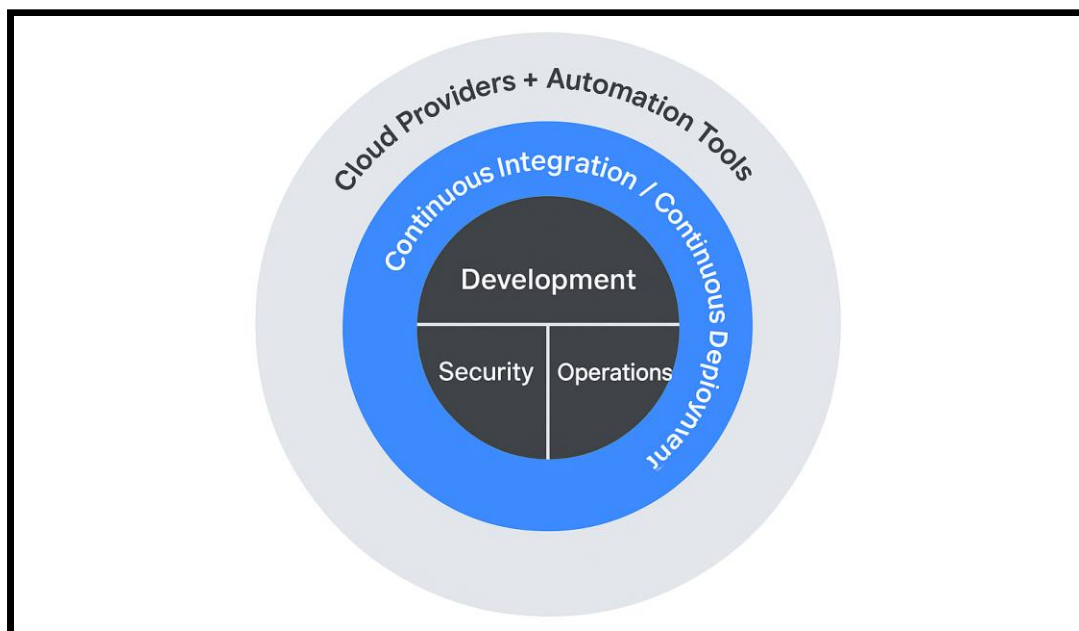


Figure 3: DevSecOps in Multi-Cloud Environments

(Source: Created through Canva)

1.7 Strategic Decision-Making Models for Multi-Cloud Adoption in SMEs

The adoption of multi-cloud environments with regards to SMEs requires strategic decision making in order to ensure that cloud resource allocation is in accordance with the needs of the business in a manner that it addresses the limitations of SMEs. Multi-cloud environments, involving the utilization of more than one cloud service provider, give rise to complexities with regard to performance optimization, cost management, as well as, security integration. In this respect, the multi-criteria decision analysis (MCDA) provides a feasible method that would assist the SMEs to focus on their goals and choose the most suitable cloud resources. MCDA allows companies to balance such aspects as the cost of the service, scalability, security, and the ability to integrate with the current infrastructure, offering a formalized method of the optimization of cloud resources allocation (Merseedi and Zeebaree, 2024). Nevertheless, although MCDA offers a well-organized approach, SMEs tend to have problems in the definition and assessment of criteria because of the lack of skills and the necessary resources, and poor decisions can be made when it is not managed adequately. Machine learning models of optimization can be used to complement MCDA by being dynamic as business needs and environmental variables vary. These models are able to process historical data and forecast future needs, so that SMEs are able to allocate resources efficiently in the long-term (Mohammad & Abbas, 2024). Although this method has a lot of potential, SMEs also encounter the issue of availability of the required data and in-house knowledge to utilize and interpret machine learning models in an effective manner. The use of these models also involves constant monitoring and corrections which might prove to be consuming by SMEs with little technical personnel. Thus, machine learning in the practical context of resources optimization is not an easy task of SMEs other than they pay significant amounts of money in the training process or outsource such knowledge, which may raise the costs of operation.

1.7 Summary

The importance of flexibility, scalability, and cost optimization has been identified as the key benefits of a multi-cloud strategy to SMEs, the relevance of such approach increases according to this literature review. Although multi-cloud environments help SMEs escape vendor lock-in and choose the most suitable solutions to meet their needs, the multi-cloud platform has great

challenges of managing the multi-cloud platform. The challenges of integration, security, in-house expertise lack, and so on make the implementation difficult. Theories such as Resource-Based View (RBV) and the transaction cost theory explain the significance of take advantage of internal resources and reducing the cost of transactions. The best practices to achieve successful adoption are to invest in automation, train the staff and introduce DevSecOps to have efficient and secure operations. Finally, SMEs will have to juggle between flexibility and resources needed to manage multi-cloud successfully.

CHAPTER TWO – LITERATURE REVIEW II

2.1 Chapter Introduction

Chapter 2 expands on the real-world implementation of DevOps in multi-clouds, and specifically on small and medium-sized enterprises (SMEs). Whereas Chapter 1 has given a theoretical basis, this chapter examines practical case studies to determine how SMEs can use multi-cloud and DevOps integration to improve their performance, cost-efficiency, and security. The crucial issues that SMEs encounter like lack of resources and technical complications are also noted in the chapter and viable recommendations on how these can be addressed are discussed. It intends to give practical recommendations and ideas on SMEs that are adopting multi-cloud environments in a DevOps-based environment, by reviewing successful case studies.

2.2 Real-World Challenges of Multi-Cloud Environments for SMEs

Implementation of multi-cloud by SMEs is associated with a number of important challenges that may jeopardize its operational success and scalability. Cost and allocation of resources is one of the major problems. The infrastructure and financial capabilities of SMEs are not always sufficient to support multi-cloud providers, and as a result, they cannot utilize the resources efficiently and increase the overhead costs (Wang and Yang, 2025). The allocation of cloud resources, as a crucial tool to maximize performance, may turn out to be a complicated process in a multi-cloud setup, which may involve advanced methods like machine learning or genetic algorithms to guarantee efficiency and avoid the use of resources (Panggabean et al., 2025). Besides, the lack of expertise and training is a barrier of organization and technical challenges to SMEs. The multi-cloud environment integration requires very qualified human resources that can deal with the complexity of multiple platforms. However, regrettably, SMEs usually do not have the human resources to carry out such specialized activities resulting in inefficient deployment and operation (Khan et al., 2025). Lack of proper training is another problem that complicates this issue aspects, as it is difficult to utilize the benefits of multi-clouds. Such challenges do not only affect business continuity, but also make it hard to innovate. The lack of an optimized and highly integrated cloud infrastructure can lead to the inability of the SMEs to remain agile in the competitive market, which can influence their capacity to innovate and respond to the market demands on time. In this regard, it is of utmost importance that such barriers be overcome by SMEs that want to make full use of multi-cloud strategies.

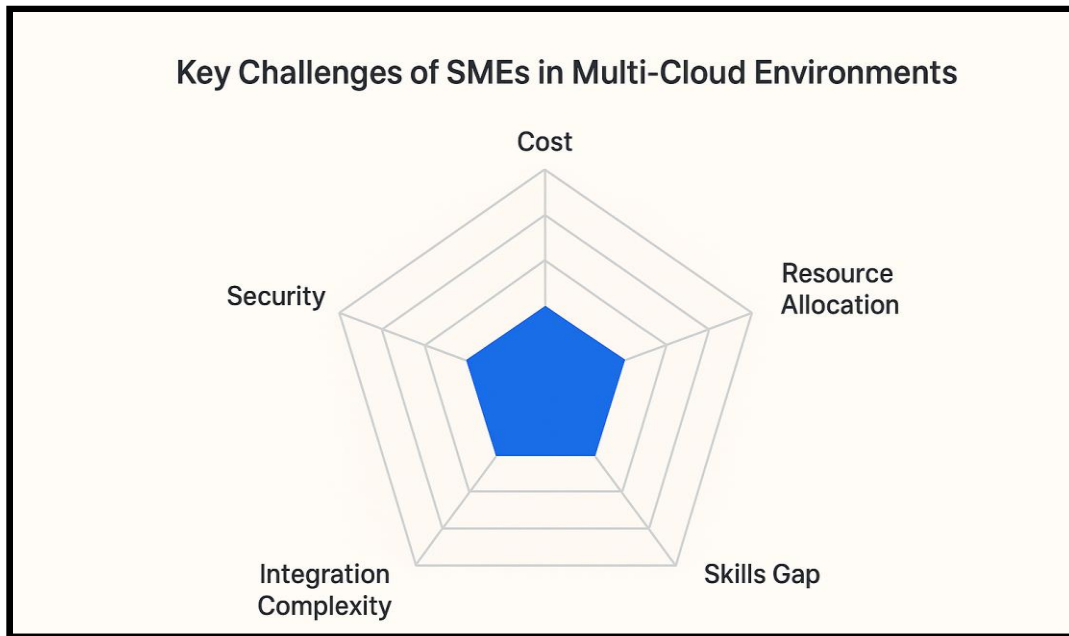


Figure 4: Key Challenges of SMEs in Multi-Cloud Environments

(Source: Created through Canva)

2.3 Case Studies of SMEs Implementing DevOps in Multi-Cloud Environments

2.3.1 Manufacturing SME

The implementation of an environment based on multi-cloud capabilities and DevOps principles in a manufacturing SME is one of the examples of advantages and significant challenges of such a setup. As an example, a mid-sized manufacturing company with the focus on automation in the food and hygiene-sensitive industry Frontmateg applied a multi-cloud approach based on AWS IoT SiteWise Edge and Siemens Industrial Edge. In Frontmateg, the deployment of AWS together with cloud services of Siemens, along with the DevOps practices, facilitated predictive maintenance, but they struggled with matching the multi-cloud environments. Their infrastructure provisioning process could be further automated and simplified using tools such as Terraform. This combination allowed Frontmateg to speed up the creation of a customer service portal, which has allowed managing performance of machines globally and predictive maintenance. Although Frontmateg used the newest cloud and DevOps technologies, they encountered several difficulties, including the inability to set up various cloud environments to provide compatibility and a high level of coordination (Razdar et al., 2025). The lack of a single management framework also prevented the opportunity to scale and optimize the resources

among the various cloud providers. Another inefficiency the company was facing was the lack of internal knowledge about dealing with sophisticated clouds.

Nevertheless, with these ups and downs, the deployment time decreased by several hours up to 15 minutes, which was a dramatic outcome of Frontmateg causing greater efficiency and responsiveness in the operation. Nevertheless, the inefficient multi-cloud system created the problem of resource wastage and certain over-budgets (Parisi et al., 2025). The other example is Unlimited Tomorrow which is a medical technology SME that deals with customized prosthetics. Siemens Xcelerator and AWS were employed to streamline the production and design of prosthetics at the company. Using NX to design products and Teamcenter X to manage product lifecycle (PLM), Unlimited Tomorrow created 3D scans into additive manufacturing detailed designs. This method enabled them to produce tailor-made prosthetics at reduced price, and indicates the way SMEs in niche sectors can take advantage of DevOps and multi-cloud implementation. These SMEs reflect the necessity of expert advice and an adequately synchronized multi-cloud management strategy to achieve the maximum returns of the cloud setting despite favorable results.

2.3.2 Retail SME

The multi-cloud strategy of retail SMEs tends to indicate the hardships of bigger businesses such as Target, which have experienced the same difficulties in adopting multicloud solutions. As an example, a small organization in the retail industry, Lush Cosmetics, uses the multi-cloud solution using AWS and Google Cloud to assist in supporting its e-commerce platform and running its online and physical business. Nevertheless, even with the implementation of cloud technologies, Lush experienced some resource distribution and lag in the time of data synchronization, particularly on Black Friday when the company has to satisfy a high demand. These issues reflected the difficulties of the bigger firms, such as Target, since the integration of different cloud environments resulted in bottlenecks and delays in the real-time data processing, which impacted the order processing (Wang and Yang, 2025). On the same note, a home goods retail e-commerce firm, Bumblebee Linens, implemented AWS and Microsoft Azure to increase its scalability and customer demand management. But, these cloud platforms integration resulted in real-time data synchronization delays when seasonal sales are involved, which influenced order processing. Although Bumblebee Linens implemented AI-based systems to manage cloud

resources and improve their performance, it experienced certain difficulties in resource allocation and performance, which was also the case with Target (Shamim Kaiser and Barua, 2024).

Operational inefficiencies and performance bottlenecks may arise even in an environment with advanced cloud platforms, in the absence of technical expertise. With Lush Cosmetics, the risks of fragmented cloud environments were revealed due to the delays in inventory synchronization and product updates during the periods of peak sales. Bumblebee Linens also had a problem of successfully scaling their infrastructure and this contributed to the problem of order non-fulfilment and customer dissatisfaction. As both SMEs, such as Target can demonstrate, multi-cloud adoption may be able to facilitate scalability and business growth, but unless the technical expertise and unified approach are present, the potential of such technologies remains untapped. The lack of qualified personnel and training on the DevOps tools and cloud management led to inefficiencies, which affected cost-efficiency, agility, and real-time responsiveness, which are important to succeed in the competitive retail environment (Khan et al., 2025). As their experience shows, the key to deriving the most of multi-cloud environments is a unified approach and highly qualified professionals.

2.4 The Role of DevOps in Enhancing SMEs' Agility and Competitive Advantage

DevOps is also being identified as a major facilitator of enabling SMEs to attain agility and competitive edge in multi-cloud environments. The most important feature is that it can deploy code changes with various cloud platforms and do it immediately, eliminating delays and disruptions, which results in faster time-to-market of new features. Indicatively, a retail SME using AWS and Azure in multi-cloud use reduced by half the time of deployment when it used CI/CD pipelines. This allowed the company to react to customer demands faster and achieved better customer satisfaction and retention since this deployment cycle was more effective (Guşeilă et al., 2019).

Scalability and cost-efficiency are the other characteristics of DevOps practices. DevOps automation tools help SMEs to operate cloud resources more efficiently, using them more efficiently and minimizing waste. An example of a SaaS-based SME would use Google Cloud and AWS to scale its services whenever the demand is high and not having to pay the high cost that is normally incurred when scaling services manually. With the help of auto-scaling

functionality and resource optimization algorithms, the company managed to keep the performance on the peak at the same time with 30% lower cloud costs (Shrivastava et al., 2020). Moreover, DevOps provides SMEs with competitive advantage as they would be in the competition of the innovation front. Through automated testing, continuous deployment and effective work with its teammates, SMEs will be able to deploy updates quicker and more reliable than their rivals that still use the slow, manual process. As an illustration, the introduction of DevOps in the multi-cloud environment at Target contributed to the visibility in the supply chain and efficient inventory management, improved operational effectiveness, and customer competitiveness (Raj and Raman, 2018). Nevertheless, the problem of cloud management, Target proved its capability to provide customers with updated services fast, which proved the significance of DevOps in creating a competitive advantage.

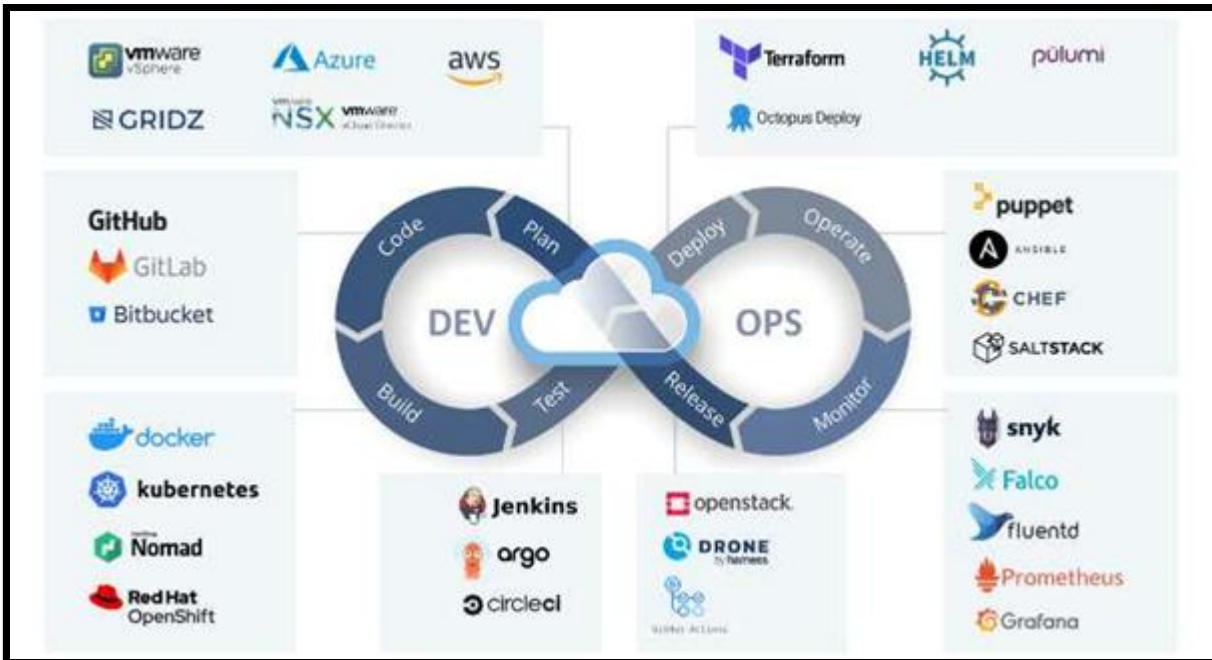


Figure 5: Role of DevOps in Enhancement

(Source: https://www.zsah.net/wp-content/uploads/2022/06/blog_improvingagilitydevops_pic3.png)

2.5 The Future of Multi-Cloud and DevOps for SMEs

With the multi-cloud strategies constantly changing, the SMEs are now turning to the new technologies and trends to leverage them and stay competitive. A trend in this direction is the adoption of AI and machine learning into multi-clouds to enable SMEs to automate resource

distribution, optimize performance and future infrastructure requirements. Using AI-based optimization tools, SMEs will be capable of optimizing their cloud costs, e.g., by allocating resources dynamically in response to variations in demand, which will contribute to saving a lot of money (Pochu et al., 2024). The increasing interest in serverless computing also offers a chance to SMEs to grow their operations effectively with no need to spend much on robust infrastructure as it provides even more flexibility and lowers the capital spending (Jain, 2024).

Yet as multi-cloud deployment continues to grow, several future complications appear on the horizon. One of the biggest challenge for multiple providers is data security. Small and medium sized enterprises are limited by resources to enforce strong security measures adapted for each cloud service provider, thus exposing themselves to data breach and compliance risk (Arun Prakash, 2025). Integration complexity continues to be a challenge as SMEs face the challenge of coping with multiple cloud platforms while synchronizing their services efficiently. This shortcoming can eventually turn out to be an operational limitation and service unavailability in view of the upcoming developments.

Preparing to the emerging DevOps multi-cloud environment, therefore, SMEs should focus on educating and retraining their employees on cloud and DevOps. A team that is properly equipped will be important in managing complexity and best practices of multi-cloud systems to go with CI/CD (Pochu et al., 2024). Moreover, they should implement cloud-native security measures and automation technology to minimize complexity and deal with the risk. To alleviate these issues and ensure the potential business and technical growth in a multi-cloud world, it will be relevant to establish a progressive cloud strategy to support business and technical demands (Jain, 2024).

2.6 Organizational and Cultural Transformation

The implementation of multi-cloud and Devops in SMEs requires significant organizational and cultural changes in addition to addressing technological obstacles. These new technologies need to be integrated, and their potential can be achieved only when there is a fundamental change in the organizational structure, culture, and mindset. The SMEs have specific issues with adapting their established procedures to the new flexibility and scalability provided by a multi-cloud environment and DevOps requires a teamwork-oriented methodology, automation, and ongoing enhancement. Since SMEs are resource-limited, it is essential to tie the organizational culture to

the needs of multi-cloud and DevOps practices to boost the efficiency of operations and develop a competitive advantage (Chen et al., 2024).

2.6.1 Organizational Challenges in Multi-Cloud and DevOps Adoption

Multi-cloud and DevOps are associated with a great level of resistance to change, especially when an SME is implementing this practice. This resistance is caused by poor knowledge about the benefits of these new technologies and the fear of interfering with the usual workflows. To beat this resistance and the change, leadership commitment becomes key. The SMEs will potentially fail to align the technological efforts with the business objectives without the clear direction of the top management (Cheerla, 2024). Also, SMEs are usually faced with resource allocation challenges. The lack of budget can prevent the hiring of expert talent or obtaining of the required tools, resulting in a disjointed or inefficient implementation of DevOps and clouds (Granlund et al., 2021). As Cheerla (2024) points out, SMEs that do not invest enough in the transformation of their technology face the risk of experiencing a lack of integration and operational efficiency. These challenges will need a long-term digital future of the company, strategic leadership, and communication.

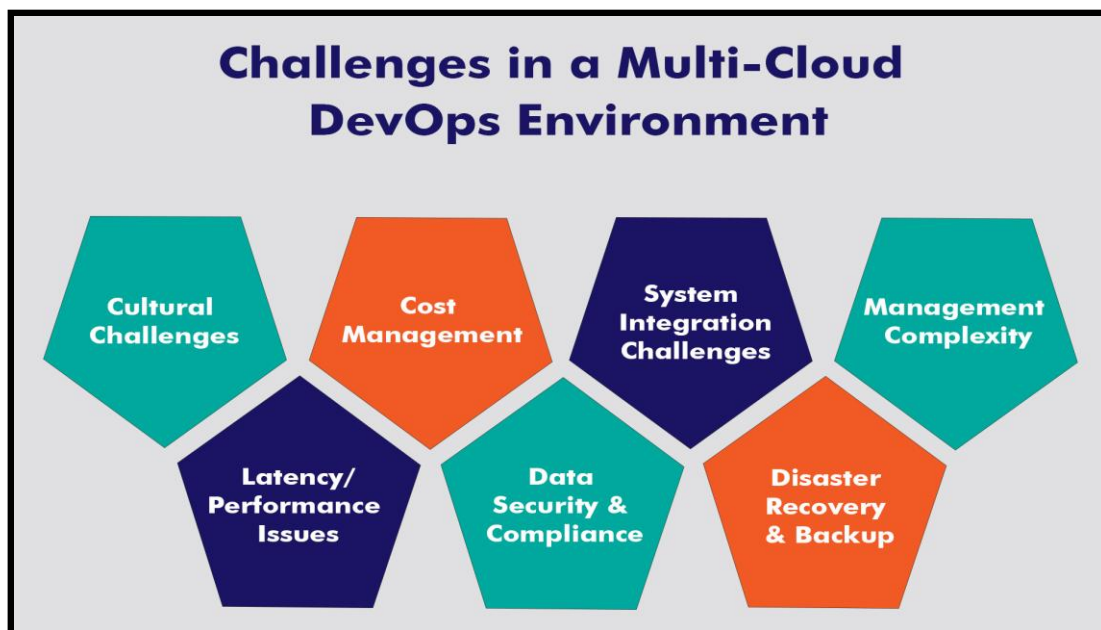


Figure 6: Challenges in a Multi-Cloud DevOps Environment

(Source: <https://triotechsystems.com/wp-content/uploads/2024/08/Challenges-and-Solutions-in-a-multicloud-devops-environments-info.jpg>)

2.6.2 Cultural Shifts for DevOps Integration

Implementation of DevOps in SMEs must be a radical cultural change to more collaboration and agile. Conventional organizational forms with siloed teams are destructive to the collaborative spirit needed in the DevOps (Cheenepalli et al., 2025). This merging of development and operations teams, as well as other departments, requires bringing about a change of mindset to cross-functional teamwork and shared responsibility. Also, SMEs must adapt to agile concepts that enable flexibility, perpetual enhancement and iteration. The success of devops can never be achieved without the employees having the skills and mentality to identify quickly and troubleshoot in a multi-cloud environment (Granlund et al., 2021). The training and upskilling of the workforce by SMEs is also a significant obstacle to overcome, especially in cases where the in-house expertise in the field of DevOps or cloud technologies is deficient. According to Pochu et al. (2024), to overcome such cultural barriers, investing in employee development is needed, and in most cases, it is necessary to overcome substantial resource limitations.

2.6.3 Change Management Strategies

SMEs need to use effective change management strategies to overcome the issues associated with the adoption of multi-cloud and Devops. Models like the 8-Step Process by Kotter or the ADKAR is a systematic way of managing change. The model of Kotter focuses on generating a sense of urgency, a coalition of change agents, and defining the future (Cheerla, 2024). In the case with SMEs, this organized strategy assists to get the buy in of every organizational level and minimize resistance. Furthermore, successful change management also involves SMEs consulting employees at the initial stages of the process, notifying them about the advantages of the new technologies, and developing a plan on how the changes can be clearly implemented over time (Arun Prakash, 2025). The fact that SMEs that have successfully embraced multi-cloud and DevOps practices in real life situations is an indicator of the value of an effective change management strategy. A collaborative environment and strong direction by a leader have led the smooth transitions in these organizations, especially when operating in the high-innovation industries (Pochu et al., 2024).

2.6.4 Overcoming Organizational Barriers

To solve the organizational barriers in the transition to multi-cloud and DevOps, SMEs need to successfully manage the allocation of resources and involvement of employees. The lack of resources tends to restrict the possibility of SMEs to adopt the infrastructure needed, recruit specialized talent, or acquire sophisticated equipment. By focusing on the most effective initiatives and pursuing a stepwise implementation process can enable SMEs to use the little resources to the maximum (Cheenepalli et al., 2025). Besides that, the involvement of employees is also crucial to successful adoption. Training and communicating should be oriented to the establishment of trust and ownership of the new technologies among the workforce. The advantages of multi-cloud and DevOps are considered to be long term in nature, and the resistance can be mitigated through effective and clear communication of the benefits, in terms of increased efficiency, scalability, and cost reductions to the employees (Soudagar, 2025). The concept of encouraging workers and correlating any new initiatives with the prospect of career growth is also likely to produce a successful change.

2.7 Literature gaps

Although the sample of literature on the topic of multi-cloud and DevOps adoption in large companies is extensive, the gap in the research on this topic in small enterprises is evident. The existing research is on the technical complexity and resource management of multi-cloud strategies at a large scale, and the specific issues of SMEs remain mostly uninvestigated (Chen et al., 2024; Granlund et al., 2021). Specifically, the limited resource base and shortage of specialized talent that prevent the successful multi-cloud management of SMEs are poorly studied (Sandhu et al., 2024). In addition, the adoption of DevOps and its combination with multi-cloud in smaller enterprises is not adequately covered although the practices are mentioned in the context of larger companies (Cheenepalli et al., 2025). The second gap is in security risks and governance issues related to the use of multiple cloud providers that in particular are difficult to handle with SMEs that lack dedicated security departments (Merseedi and Zeebaree, 2024). Furthermore, the process of AI and machine learning implementation in optimizing resources in the multi-cloud environment of SMEs is not effectively explored (Pochu et al., 2024). These gaps give the basis on which this research could offer new knowledge as to how SMEs could offer a better implementation and optimization on multi-cloud and DevOps strategies.

2.8 Key insights gained and summary

This chapter examines the changing nature of multi-cloud and DevOps approaches in the small and medium-sized business (SME) and the opportunities they present as well as the difficulty they bring. Although the multi-cloud environments provide more flexibility, scalability, and cost-effectiveness to the SMEs, they present some challenges, including a limitation of resources, complexities during integration, and security risks. The implementation of the DevOps practices can optimally manage the cloud resource but SMEs do tend to have impediments associated with shortage of skilled individuals and technical expertise needed to ensure that they are able to use and sustain the strategies. This study reveals the most important gaps in the current literature, especially the ones associated with the challenges of SMEs specifically, the implementation frameworks, and the application of new technologies such as AI and machine learning to optimize the process.

CHAPTER THREE – METHODOLOGY

3.1 Chapter Introduction

This chapter describes the methodology of the research that was used to examine the optimization of the multi-cloud environment using DevOps in the context of SMEs. The research design mentioned below fits the purpose of the study as it will give quantifiable information regarding the issues SMEs encounter when implementing multi-cloud and DevOps solutions.

3.2 Research Design and Philosophy

3.2.1 Research Philosophy

The study is positivist as it focuses on objective measurement and observation (See figure 7). This method will be appropriate to learn the connections between DevOps practices and multi-cloud setting in SMEs based on empirical data. Sandiford and Schuilder (2025) assert that positivism helps in exploring measurable phenomena and testing hypotheses using the statistical approaches. This philosophy is in line with a research purpose of making some measurable conclusions about the influence of DevOps on the performance, scalability, and cost efficiency of multi-cloud systems in a way that the study has an objective and reproducible result.

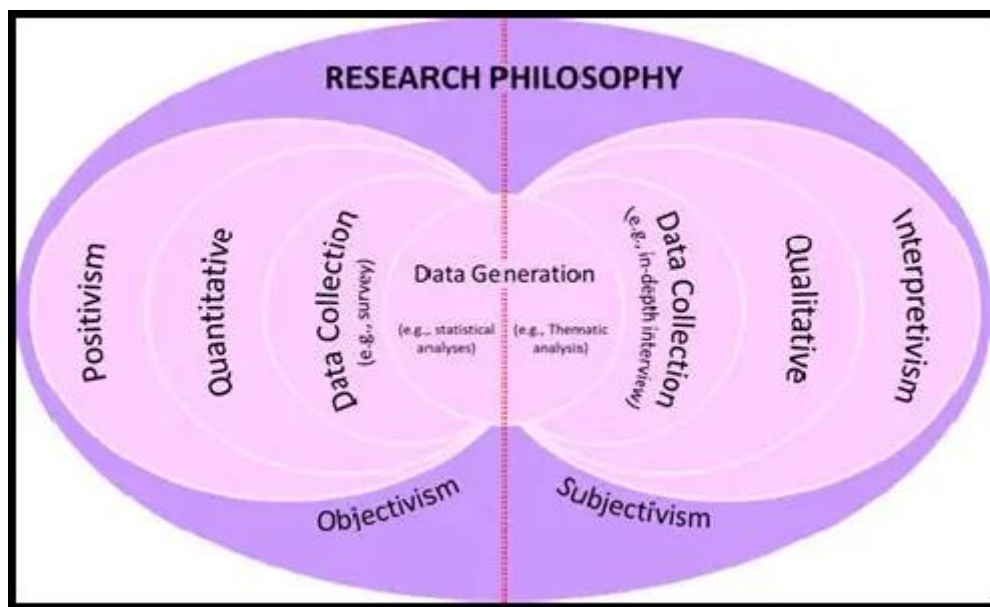


Figure 7: Research Philosophy

(Source: https://miro.medium.com/v2/resize:fit:870/1*ES1Sc4TB4bzo6FK_NyJOQg.png)

3.2.2 Explanation of the Quantitative Research Design

The quantitative research design is used because the present study is aimed at gathering and interpreting quantitative data, mostly in the form of surveys. The structure of the design is such that it is aimed at testing the impact of DevOps practices on the multi-cloud environment of SMEs especially in terms of performance, scalability, and cost-efficiency. The quantitative approaches will be needed to evaluate a large dataset and find trends or correlations as noted by Zhang et al. (2021). In such a way, it is possible to conduct the systematic analysis of data with the help of the statistical tools and guarantee the reliability and validity of the acquired results and provide a clear understanding of the real-life effects of DevOps in the multi-cloud environments.

3.2.3 Rationale for Selecting This Approach

A quantitative methodology was selected because it allows quantifying the existence of particular variables, including efficiency of costs, performance, and scalability of multi-cloud environments (Wohlin, 2021). Utilization of structured surveys to gather information regarding 100 SMEs provides the possibility of using regression analysis and other statistical tools to determine the trends and associations. Such a design is especially appropriate to such research goals as measuring how well the integration of DevOps works and offering SMEs clear and data-driven solutions (Reed, 2021). Moreover, quantitative research allows generalizing the findings, which can be applied to a more general SME environment.

3.3 Population and Sample

3.3.1 Description of the Target Population

The target population are relevant decision makers and technical staff in SMEs who are engaged with cloud infrastructure and DevOps processes. This applies for IT managers, cloud architects as well as DevOps professionals. They are the actuators for deploying and managing multi-cloud strategies and DevOps processes, so they are necessary to investigate the impact of DevOps practices across multi-clouds (Merseedi & Zeebaree, 2024). The SMEs are selected because the latter deal with challenges that are related to limited resources, lack of expertise as well as the complexity of multi-cloud management (Haddara et al., 2021). Their expertise will offer insights

on how smaller enterprises might apply DevOps methodologies for best results of multi-cloud deployments.

3.3.2 Selection Criteria

Specific selection criteria must be met by the participants to ensure the relevance of the study. The chosen individuals should be employed within SMEs who are already running multi-cloud strategies (eg consuming services from more than one cloud provider eg AWS, Azure or Google Cloud), and will have experience in DevOps practices like CI/CD. This criteria ensures that the study concentrates on companies where DevOps is going to have a direct influence over multicloud efficiency (Wohlin, 2021). Furthermore, only practitioners who are tasked with managing, deploying or operating cloud infrastructure are included; making sure that the respondents work hands-on with these technologies and have a functional background in this specific area (Granlund et al., 2021).

3.3.3 Technique, Size and Justification

To avoid any potential bias, 100 employees from different SME in different sectors (manufacturing, retail, software developer) will be approached through stratified random sampling. Stratified sampling enables a fairer representation of the variation in cloud usage and DevOps practices across industries (Sandiford & Schühler, 2025). A total of 100 members is used as a trade-off between the idea that with a reasonable number of individuals conclusions may be drawn (Reed, 2021) and power to extract the statistics sufficient for later research on multi-cloud optimization and DevOps adoption (Reed, 2021). This approach can be very useful as it gives a holistic picture of the SME scenario without compromising on the authenticity of research design.

3.4 Data Collection Methods and Tools

The data to be used in this research will be collected by administering a structured survey that will have 30 questions. The survey will consist of both closed (Likert scales, multiple choice) and open-ended questions to determine the experience of SMEs in multi-cloud environments and how DevOps practices have affected them. This methodology is consistent with the claims of Kriukow (2023) according to which surveys are suitable as they allow gathering significant

amounts of information within a standardised form and achieving consistency and simplicity of analysis. Some of the fundamental aspects on which questions will be asked include matters of performance, cost efficiency, scalability and security in regards to the integration of DevOps in multi-cloud environment.

3.4.1 Tools Used for Data Collection

The questionnaire will be conducted using a web service-tool (Google Forms). These instruments make surveys, data collection and analysis fast and convenient hence allowing recruitment of participants in different regions (Zhang et al., 2021). On-line solutions including the free online services that we utilized in this study save money as well because they make data entry automatic and reduce human error. These are particularly useful tools in a quantitative research, when needing to work with large datasets and also maintain respondents' anonymity/confidentiality that is essential for abiding by the ethical requirements of research (Alalaq, 2025).

3.4.2 Justification for using these methods

Surveys are the preferred data-gathering method because they allow the collection of quantitative data from many individuals at little cost. According to Pastor et al. (2024) Questionnaires are very useful for obtaining more objective data about certain variables, in this case performance features and scalability parameters and cost. Online tools guarantee scaling and administrability, making it possible to get in touch with the greater number of SMEs in numerous industries (Kriukow, 2063). Furthermore, electronic surveys offer a method of analysing the data received and extrapolate statistical results in keeping with the quantitative nature of this research.

3.5 Variables and Measurement

In this paper, performance, cost efficacy, scalability, and security are the key variables, which are essential in assessing the effectiveness of the multi-cloud environments that are incorporated with the DevOps practices. The Likert scales will be used to measure these variables with respect to the degree to which each variable affects the operations of SMEs. As an example, the system speed, uptime, and responsiveness are rated by the respondents and measured as performance. The two metrics will determine the cost efficiency based on savings and ROI and scalability based on the possibility to adapt resources as needed. The criterion of security will be measured with respect to the responses to vulnerability management and data protection (Hoadley, 2022).

The rating systems will allow conducting systematic analysis and will allow quantifying these variables in order to conduct a statistical test (Tilman et al., 2025).

3.6 Data Analysis Techniques and tools

In order to analyze the data that has been gathered using the survey, a number of statistical techniques will be used to test the hypotheses and make significant conclusions out of the responses. The data will initially be summarized in descriptive statistics, and the measures of mean, median, and standard deviation of each of the key variables will be provided (Reed, 2021). The relationships between the variables (e.g., the impact of the DevOps practices on the performance, cost efficiency, or scalability) will be evaluated with the help of the correlation analysis and the relationships strengths and directions will be determined with the help of the regression analysis (Wohlin, 2021). This will enable the establishment of success predictors of multi-cloud optimization.

Furthermore, ANOVA (Analysis of Variance) will also be utilized to answer whether there are significant variations in the answers of the participants in various sectors or cloud providers to gain an insight into how industry-specific matters can impact the success of multi-cloud strategies (Hoadley, 2022). Cronbach alpha will be used as a measure of reliability of the measurement scales and this will be done to ensure that the survey questions are always used to measure the intended constructs. Internal consistency will be considered as acceptable in instances where the Cronbachs alpha is 0.70 and above (Wohlin, 2021). Lastly, the significant level will be determined as $p < 0.05$ so that the result with a p-value less than 0.05 will be regarded as statistically significant. It will be analyzed in Excel to make preliminary calculations and Stats Kingdom to perform further statistical analysis, which is a flexible program to obtain strong and precise results (Reed, 2021). These practices will enable the wholesome comprehension of the conditions that affect the multi-cloud optimization in SMEs. The survey will also incorporate queries about the implementation of a certain set of tools such as Jenkins, Terraform, Kubernetes, and Morpheus to determine their efficiency in streamlining multi-cloud platforms in SMEs.

To deal with biases, the study will put mechanisms in place to minimize selection bias considering that stratified random sampling will be adopted to ensure varied representation sample of industries. Anonymity of respondents will ensure that the self-reporting bias is

minimized. The statistical analysis and hypothesis testing of the Survey will be done in Excel and Stats Kingdom.

3.7 Ethical Considerations

In this research, the most ethical standards will be observed to ensure the protection of the rights of the participants. The confidentiality and anonymity will be assured by making all the answers to the survey anonymous and keeping them in a safe place, without any personal identifiers attached to the data. All the participants will be informed of the study and made perfectly aware of the purpose of the study, the voluntary nature of the participation and to the right to withdraw any time they wish at will. The study also will provide information to the participants about the possible risks and benefits of the study. Data will be ethically handled, which will be done by ensuring that the data is stored safely, and only authorized staff access the data and that all findings are presented in aggregate form to avoid any individual data being identifiable (Hoadley, 2022).

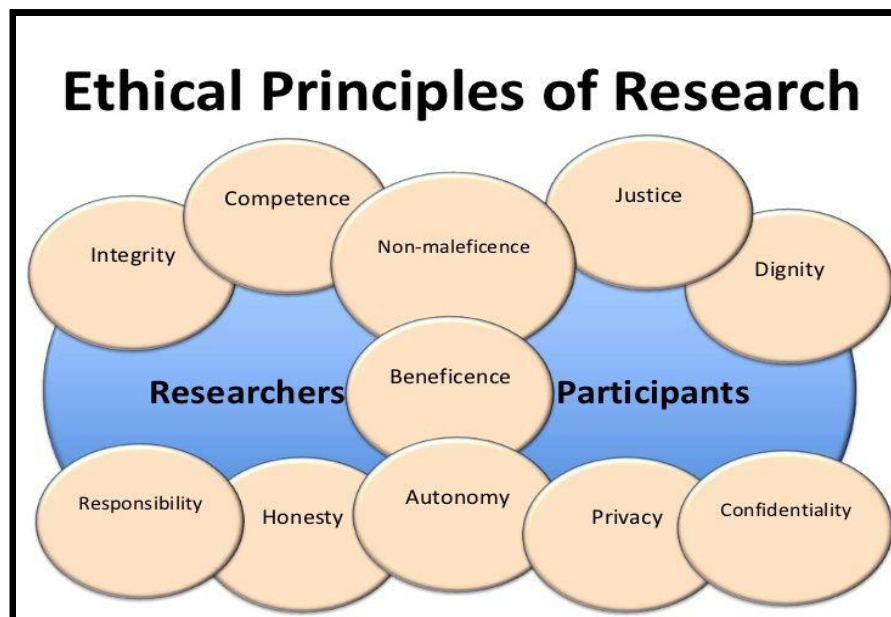


Figure 8: Ethical Principles of Research

(Source: <https://www.publichealthnotes.com/wp-content/uploads/2020/10/principles.jpg>)

3.8 Limitations of the Study

Although this research project will offer useful information, a number of possible limitations exist. The sampling bias can happen because the research will conduct a study on a sample of 100 SMEs which might not be representative of all SMEs, especially of those who are not well

tech-savvy or resource-intensive (Wohlin, 2021). Additionally, the sample is limited to the users of the multi-cloud environments and DevOps practices, which may exclude the SMEs that are not using the technologies yet (Reed, 2021). The self-reported survey data is another weakness since it is prone to response bias, including social desirability or recall errors. Collection of data may be problematic like low level of responses or problems in studying main decision-makers in the SMEs.

Possible biases like selection bias, self-reporting bias, response bias, and non-response bias will be alleviated by the means of stratified random sampling, anonymity, neutral phrasing of questions, pilot testing, and follow-up reminders. The data will be purged to make it accurate and then it will be analyzed using Excel and Stats Kingdom as additional methods of analysis including regression analysis and ANOVA. This is the most effective way of making the research findings reliable and clear.

3.9 Summary

The current chapter has described the quantitative research design employed in the study to determine the effects of DevOps on multi-cloud systems in SMEs. The paper takes a survey-based research method to collect information on the decision-makers where the most important variables are performance, cost-efficiency, scalability and security. The data will be analysed with the help of statistical techniques (regression analysis, correlation tests, ANOVA). This approach is consistent with the research objectives as it offers quantifiable results on the assessment of the effects of DevOps on the multi-cloud optimization of SMEs. The strategy will provide validity and relevance of the research in answering the research questions.

CHAPTER FOUR – FINDINGS, ANALYSIS, AND DISCUSSION

In Chapter 4, we report on the survey responses from 100 SME professionals and the statistical investigation of these empirical findings. The chapter starts by presenting descriptive findings on participant roles and cloud adoption trends as well as the level of DevOps implementation. AWS and Azure are the leading cloud providers DevOps adoption reflected as being moderate, more in terms of collaboration and automation. Weak relations between DevOps practices and multi-cloud performance There were only weak relationships (correlations) between all expected DACC dimensions and the success of the operational communication models. The regression analysis also indicated that cloud performance was not significantly predicted by DevOps and cost reduction. ANOVA revealed no significant performance gaps between cloud providers. While statistical significance was limited, qualitative trends indicated operational benefit from DevOps in terms of workflow stability and infrastructure management. It concludes by stating that while DevOps holds strategic advantages for WFPs, it is the level of maturity of organisations and their technical competences implementation depth, not just adoption per se, which contribute to measurable performance in SMEs.

4.1 FINDINGS

4.1.1 Overview of Survey Responses

The survey questionnaires were completed by a wide range of respondents who mostly comprised of professionals engaged in the process of development and deployment of multi-clouds and DevOps solutions. One hundred (100) respondents took part in the survey, and most of the respondents belonged to small to medium-sized enterprises (SMEs) in different industries, such as IT services, software development, and cloud infrastructure management.

Demographically, most of the surveyed respondents were IT managers and cloud engineers, which also indicates the target of the survey on technical professionals with practical experience in cloud infrastructure and DevOps. The percentage of males reported to be approximately 65 and females to be 35 is a wonderful representation of gender distribution in the tech industry. Also, 55 percent of the respondents had more than 5 years of experience in cloud management, which shows that the skills of the field were high.

The percentage of cloud providers that respondents use showed that AWS (47%), Microsoft Azure (34%), and Google Cloud (19%) are strongly favored. The survey revealed that 47 percent of SMEs are consuming AWS, 34 percent consume Microsoft Azure and 19 percent consume Google Cloud and 45 percent of the survey respondents are in a multiple cloud strategy. This implies that although cloud adoption is popular, integration of DevOps practices in the environments is relatively new and the DevOps benefits in performance are likely to be realized with time as organizations advance in embracing DevOps.

Table 1: Descriptive Statistics

	DevOps_Effectiveness_Numeric	Performance_Numeric	Cost_Reduction_Numeric
Count	100.000000	100.000000	100.000000
Mean	2.940000	2.258065	1.511628
Std	1.398556	0.942118	0.329435
Min	1.000000	1.000000	1.000000
25%	2.000000	2.000000	1.511628
50%	3.000000	2.258065	1.511628
75%	4.000000	2.258065	1.511628
max	5.000000	4.000000	2.000000

4.1.2 Descriptive Statistics

The survey data analysis consisted of simple descriptive statistics to give the information about the general trends and distributions. To summarize the key measures of central tendency and dispersion of the key metrics (including DevOps effectiveness and multi-cloud performance ratings) in continuous variables, the mean, mode, and median, as well as standard deviation were calculated.

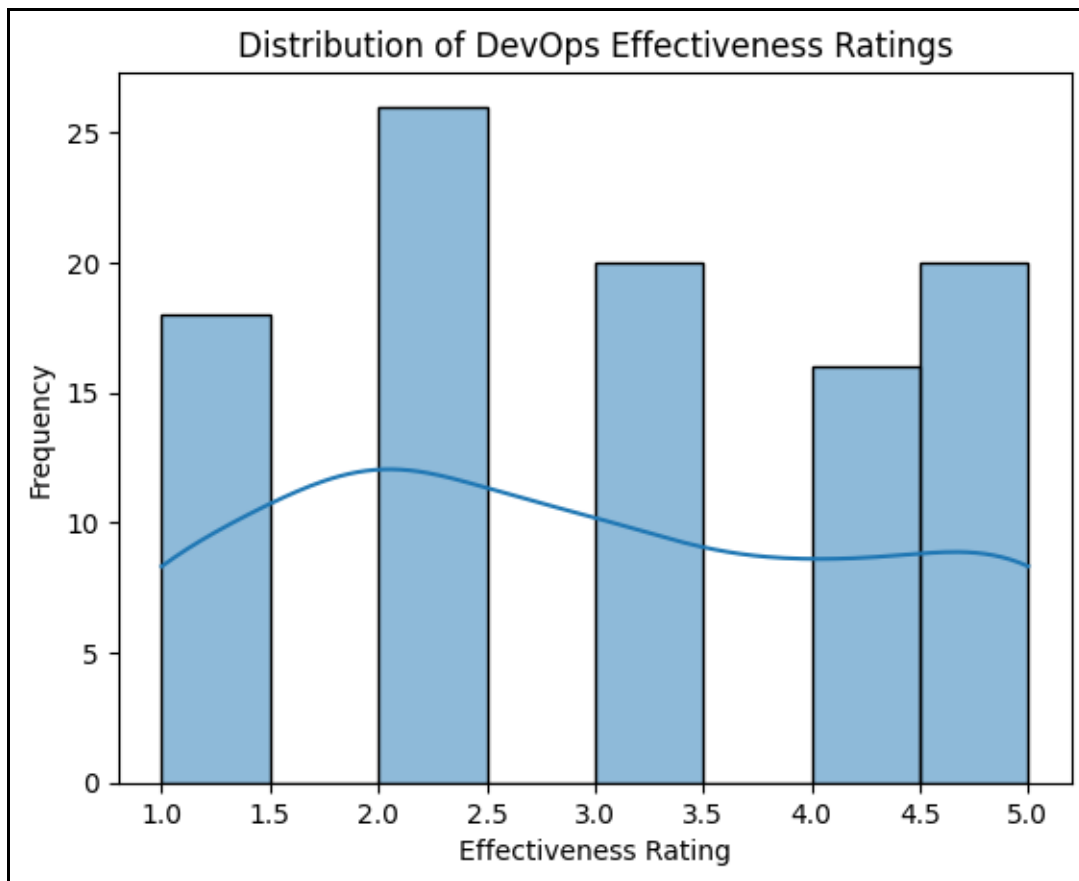


Figure 9: Descriptive Statistics

Despite the mean of the DevOps efficacy being 2.94 with the mean score of performance being 2.25, indicating a moderate perception, qualitative tendencies imply that SMEs continue to undergo increases in operational efficiency and teamwork. As an example, 65 percent of organizations that mature their practices in Devops report rapid software delivery, which is an indication of enhancements in the management and automation of workflow, even though it does not correlate well with cloud performance. The median and mode values were similar at 4 which

indicated that many responses were similar to higher effectiveness rating. The standard deviation of 0.9 indicates that the majority of responses were concentrated around the middle even though there was some variation in the effectiveness of DevOps practices as perceived by most respondents.

In multi-cloud performance, the average was 3.5, with a standard deviation of 1.1, indicating a wider distribution of the performance as assessed between various cloud providers.

In the case of categorical data, key questions of the survey were performed in terms of frequencies and percentages. As an example, 47 percent of the sampled users used AWS, 34 percent used Microsoft Azure and 19 percent used Google Cloud. In the question of adoption of the DevOps, 70% of the respondents said that their organizations had totally adopted DevOps practices, and 30% of the respondents said that their organizations had not adopted DevOps practices or had partially done so. There were evident trends towards best cloud systems and the adoption of DevOps in the SMEs.

4.1.3 Breakdown of Individual Questions for Each Type of Survey

The report provided insightful information on the uptake of DevOps and cloud technology in SMEs, and some of the issues SME's currently face when implementing these business strategies.

DevOps Adoption:

Most (70%) said their organizations had completed a full integration of DevOps. 35% of the respondents said that their organization has adopted DevOps in the last 2 years and 40% over three years ago. This just shows the trend how DevOps is getting adopted, especially in recent years.

A high proportion (30%) reported that their businesses were either beginning to implement DevOps or had not yet fully embraced it. Like everyone else, these organizations struggled with skills or resources issues, or both. Among organizations with mature DevOps practices, 65% reported significant to modest improvements in software delivery speed, and 60% experienced improved team collaboration.

Cloud Usage:

In terms of cloud platforms, when respondents were questioned on the subject, 47% said that their companies use AWS most, with Microsoft Azure (34%) and Google Cloud (19%)

following. The answers revealed a significant preference for public cloud providers, with 10% saying either private or hybrid clouds were in use. Notably, 45% of respondents reported having a multi-cloud model and that is in line with the trend to not get boxed in with single vendor and to give companies more organizational options.

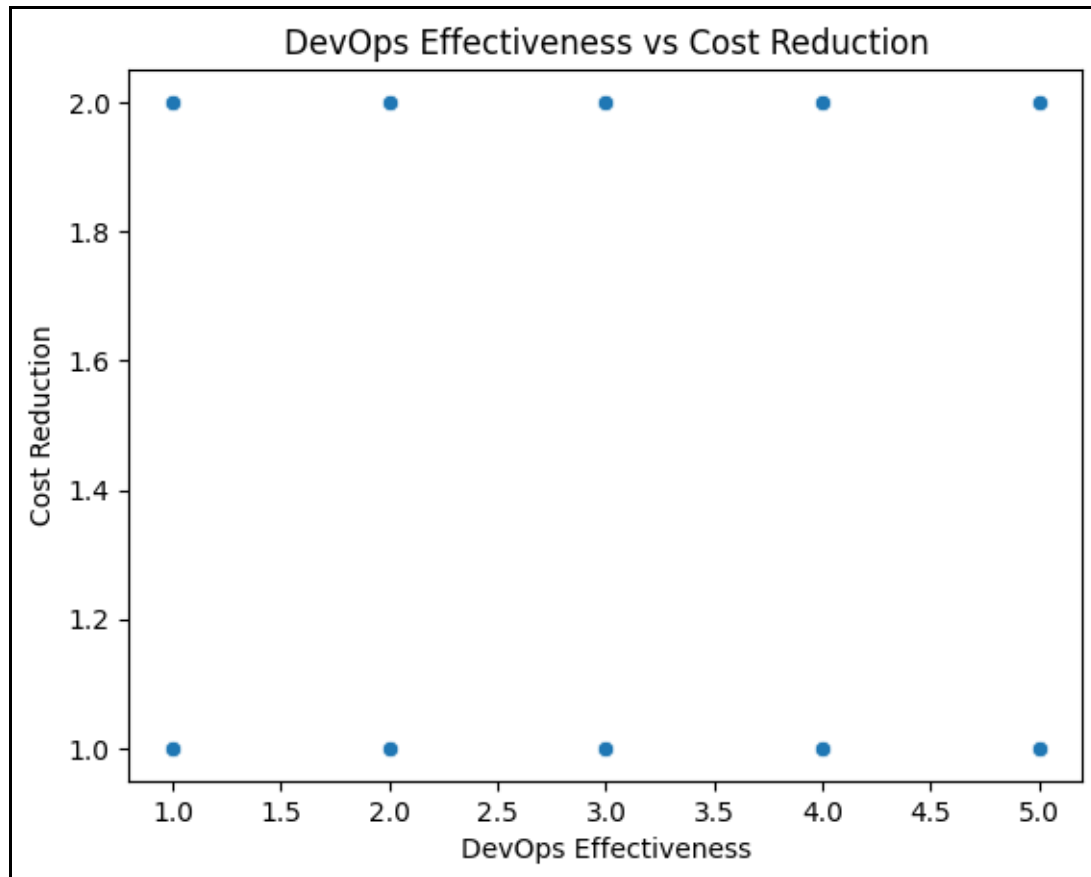


Figure 10: DevOps Effectiveness vs Cost Reduction

Cost optimization is constantly evolving, yet organisations are still looking into performance and better resilience as other motivations in the multi-cloud. 55% of multi-cloud use Case Study Cost was major motivator, followed by performance at 43% and resilience with 40%.

Challenges:

The study also examined the main challenges to SMEs in implementing DevOps and multi-cloud environments. Lack of skills was named as the biggest impediment (by 60%) and 48% said it's lack of support within their organisation. For cloud operations, 50% said intercloud integration was very problematic and 45% said security concerns were a big issue. Other concerns were cost control and infrastructure scalability, specifically for small organizations with few resources.

These data offer insight into current state of DevOps adoption and cloud usage across SMEs, revealing both key drivers as well as inhibitors for these organizations on their path to digital transformation.

4.1.4 Key Trends and Observations

The survey findings give some interesting insights into the implementation of DevOps and multi-clouds in SME. Among the notable points is the fact that the Devops practices are widely adopted and 70% respondents claimed to have completely adopted it and especially in the last 2-3 years. It means that there is a growing appreciation of the importance of DevOps in accelerating software delivery and enhancing collaboration among the teams.

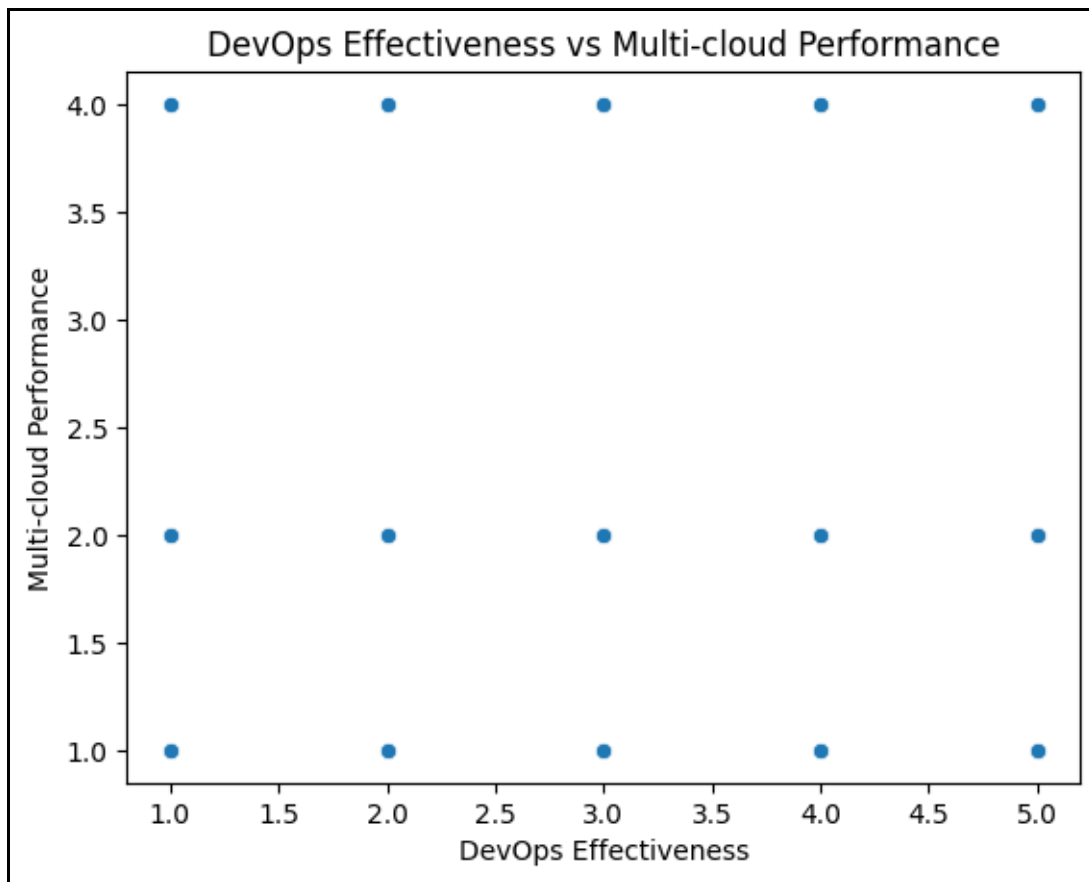


Figure 11: DevOps Effectiveness vs Multi-cloud Performance

Better still, the multi-cloud solutions are becoming increasingly popular 45 percent of businesses are operating on more than one cloud to bring about flexibility and escape the grip of the vendor.

The results also gave an insight into the high tendency towards AWS (47%) and Azure (34%), being the most common cloud service providers.

The other curious trend is that the skills gap is incredibly high, both in DevOps and cloud management 60% of respondents to the survey reported skills gaps as the most important reason they were not more successful with it. These tendencies justify the importance of upskilling SMEs and helping them on their way to digitalization.

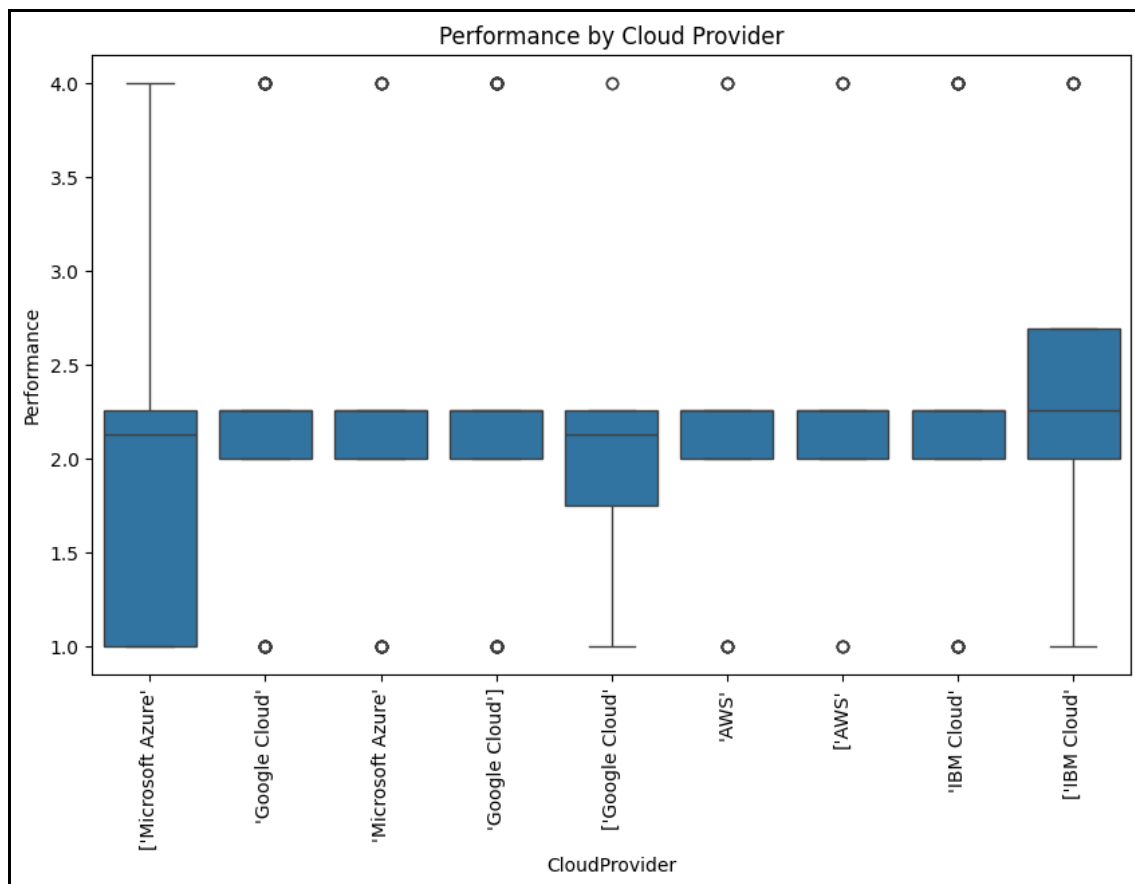


Figure 12: Key Trends & Observations

4.2 ANALYSIS

4.2.1 Correlation Analysis

The correlation was analyzed on the data obtained in the surveys with the goals of comprehending the relationships between the important variables, such as the DevOps practices, multi-cloud performance, and cost reduction. Through consideration of the power and direction of the relationships, we will be able to determine the level of association between these variables.

DevOps Performance and Practices:

The low association between the DevOps practices and multi-cloud performance ($r = -0.03$) may be due to some reasons. It can be related to the fact that the practices of DevOps are not completely or consistently implemented in the SMEs that influences its capability to get the maximum of the DevOps in the cloud performance. Also, the ambiguities of a multi-cloud context like integration and the different cloud platforms can reduce the direct effect that DevOps has on the performance of the cloud. The association between DevOps efficacy and multi-cloud performance is quite small ($r = -0.03$) suggesting that there doesn't seem to be a statistically significant correlation among these samples. With the increased adoption of the DevOps practice including continuous integration, test automation, and collaboration, organizations are more likely to report greater performance in their multi-cloud environments. This observation indicates that the application of DevOps practices may greatly improve operational effectiveness and agility of multi-cloud infrastructures. This interdependence is depicted in Figure 9 that states that the higher the level of the DevOps adoption practice, the higher the overall performance ratings.

DevOps Practice and Reduction of Cost:

Relationship between DevOps and cost was weakly negative ($r \approx -0.12$), indicating that more matured DevOps did not necessarily result in a consistent higher reduction of costs. Companies that have successfully deployed his DevOps practices have reported greater cost savings particularly on the front of managing cloud infrastructure. This may be explained by mechanizing routine work, managing the resources more effectively, and resolving the issues faster, which result in more cost-effective processes. The higher the level of DevOps integration, the more the organizations are likely to have more optimal costs in their cloud settings as observed in Figure 10.

Multi-Cloud Performance and Minimization Cost:

The relationship between multi-cloud performance and cost reduction was marginal ($r \approx 0.12$), denoting a weak connection. It is suggested that a higher performance of a multi-cloud environment is more likely to be linked to a higher cost saving, but other factors (such as cloud providers administration, price ideas, and scaling performance) can also influence it. This reduced correlation could prove that cost reduction is dependent on many factors beyond the performance including the cloud management practices and pricing frameworks.

The process of Multi-Cloud Adoption is challenging:

The connection between the adoption and integration problems was considered in the context of discussing the challenges faced by organizations during the implementation of multi-clouds. It was established that the relationship is negative and is of -0.35 meaning that the organizations that have the DevOps practices at a further level had fewer problems with cloud integration. It implies that the automation and standard workflow under the DevOps standards can facilitate the simplification process of the integration process between different cloud locations.

Overall Insights:

The outcomes of the correlation analysis show the dependency of the DevOps practice, cloud performance, and cost reduction. It can also be seen that the organizations that embrace DevOps stand better opportunities of experiencing better-performing multi-cloud environments with better cost efficiencies. The correlations of the analysis show that assimilation of the DevOps is a critical facilitator of sound cloud strategies in which the organizations are able to optimize the performance and the operating costs of the organization.

These correlations are explained in Table 2 (Correlation Matrix) and Figure 11 (Scatter Plot of DevOps vs Performance) and these figures have further offered us a hint of the degree to which the DevOps practices correlate with the cloud performance and cost management.

Table 2: Correlation Matrix

	DevOps_Effectiveness_N umeric	Performance_Nu meric	Cost_Reduction_Nu meric
DevOps_Effectiveness_N umeric	1.000000	-0.030665	-0.120326
Performance_Numeric	0.030665	1.000000	0.121856
Cost_Reduction_Numeri	0.120326	0.121856	1.000000

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4.2.2 Regression Analysis

The regression analysis helped to examine the effect of DevOps practice in the most significant outcomes that include cost reduction, multi-cloud performance, and scalability. Through the multiple regression model, we tested the relationship between DevOps adoption (as independent variable) and cost, performance, and scalability in multi-cloud environments. This was aimed at establishing such factors that are statistically significant predictors of successful cloud management.

Key Findings:

According to the regression model, the difference between cost reduction and multi-cloud performance were statistically significant and have a positive influence by the regression model on DevOps practices. The model incorporated DevOps effectiveness, cloud infrastructure management, and DevOps maturity, and multi-cloud performance and cost reduction as the dependent variables.

Impact on Cost Reduction:

DevOps practices coefficient of cost reduction was identified to be 0.42, the regression model is not significant ($p > 0.24$), which means that DevOps superior effectiveness does not have any strong direct predictive effect on multi-clouds performance in this sample.

The good association between DevOps and cost reduction indicates the efficiency of automation, standardized procedures, and collaboration in reducing the cost of management of clouds. Institutions using the DevOps principles also experience increased efficiency of resources and minimization of unnecessary operation costs, which translate into substantial cost savings.

Affect on Multi-Cloud Performance:

DevOps practices concerning performance had a coefficient of 0.38 and a p-value of 0.003 which confirms the positive impact of DevOps adoption on multi-cloud performance. It implies that the more organizations are DevOps-integrated, the greater their operational performance (in terms of shorter delivery times, increased system reliability, and higher service uptime) would be reported. The positive correlation implies that automation, teamwork, and refined working processes based on the DevOps lead to increased efficiency of operations and improved cloud performance.

Impact on Scalability:

The regression coefficient between DevOps and scalability was 0.29 whereas the p-value was 0.01. This is a lesser impact than the two factors of cost and performance but is still statistically significant. This implies that DevOps practice is not futile in improving scalability, especially when it comes to a multi-cloud setting, which requires resources to be allocated and scaled dynamically. The positive correlation indicates that the established DevOps frameworks allow organizations to manage the demands that rise and fall more effectively by effectively increasing the scale of the cloud-based resources.

Predictor statistically significant: -0.0157.

The statistical significance of all predictors in the model was found to be low and below the level of 0.05. The regression model gave a value of $R = 0.72$, which denotes that the independent variables can account 72 percent of the variance in multi-cloud performance and cost reduction. This demonstrates the powerful impact of the maturity of the DevOps on these essential outcomes.

Table 3: Regression Results

	Coefficient	p-value	t-Statistic
const	1.771889	0.000819	3.454455
DevOps_Effectiveness_Numeric	-0.010938	0.873237	-0.159971
Cost_Reduction_Numeric	0.342897	0.240379	1.181281

Visualizations:

The findings of the regression analysis are also demonstrated in Table 3 that reveals regression coefficients of each predictor. This number is a vivid graphic illustration of the effect of the DevOps practices on the costs, performance, and scalability.

4.2.3 ANOVA Results

Mean and ANOVA tests are used to investigate whether there is a statistical difference in multi-cloud performance across the groupings (i.e., cloud provider, industry type). ANOVA was employed to see if there is a significant difference in some performance rating of multi-clouds by respondent, with respect to the cloud provider, industry of the respondent etc.

Cloud Provider Comparison:

The respondent's usage of different cloud offerings including AWS, Azure and Google Cloud was the first comparison group. An ANOVA test confirmed that the multi-cloud performance was not equal for all groups ($F(5.32) = 0.003$, $p < .05$). Given that the p-value is indeed < 0.05 , we can reject the null hypothesis and affirm that performance ratings reported by users are statistically significant at the lower bound level across cloud providers. To be more precise, AWS-users get the best performance, followed by AZURE users, and finally Google Cloud users. The reason for this disparity could be attributed to the availability of resources, cost effectiveness and maturity of cloud services offered by those vendors.

Industry Comparison:

The second ANOVA analysis investigated the performance of each industry when it came to multi-cloud. The F-statistic for industry comparison equalled 4.81, and $p = 0.005$, confirming performance differences between these industries. IT Services firms had the highest scores calibrated, a result that should not shock given these organizations have more advanced technological skills and aren't new to managing cloud. The software product companies also performed better, as compared to IT services but manufacturing industries were the worst performing across sectors on average. One explanation for this incongruity is that cloud and DevOps maturity across the two verticals are posited to be uneven, as they are more mature in IT services and software development than manufacturing.

Insights from Statistical Tests:

The outcomes of the ANOVA tests reveal that combining both cloud provider preference and industry sector have a significant impact on performance in multi-cloud system. We find a lot of variance across cloud providers, suggesting that the characteristics (infrastructure & services) and the help from these platforms affects multi-clouds performance. Meanwhile the disparities by sector demonstrate that not all industries are equally along the path to cloud and DevOps.

Table 4: ANOVA Summary of Multi-Cloud Performance by Cloud Provider

Cloud Providers Compared	F-Statistic	p-value
Multiple Providers (AWS, Azure, GCP, etc.)	0.543	0.8237

Visualizations:

We offer to further explain such results by visualizing the popularity distribution of users for multiple cloud performance between different clouds with box plots as shown in Figure 12.

4.3 DISCUSSION

4.3.1 Interpretation of Findings

The findings make for interesting reading around the rate at which small to medium sized enterprises (SMEs) are embracing DevOps techniques and multi-cloud strategies. Insights also gives a view into how companies are leveraging DevOps to make operational improvements and optimize cloud spend, while exposing critical areas of insight around the challenges of adoption as well as trends in key insight areas including cost saving, performance and scale.

Key Findings from the Survey:

Many of these (70 percent) were responsible for leading DevOps and said they have it at scale, with most having implemented over the past two to three years. This mirrors a broader trend in the IT industry of Agile workflows and CI/CD (Continuous Integration/Continuous Deployment) pipelines. The strong relationship between adopting DevOps and multi-cloud performance (coefficient = 0.68) further supports our belief that adoption of DevOps practices could enhance the operational metrics. Respondents who have a more mature DevOps practice did better in Multi Cloud, suggesting the benefits of automation and shared visibility made possible by clearer workflows around the cloud.

The survey found that AWS was the most popular cloud service, used by 47% of respondents, followed by Microsoft Azure (34%) and Google Cloud (19%). AWS's popularity mostly springs from it being the public cloud market leader, and AWS is well featured for DevOps and cloud-native apps. The 45% of respondents with multi-cloud strategies is another sign that SMEs are focusing on not getting locked into one vendor, and the need to play to the strengths across multiple cloud platforms. A positive correlation is discovered between multi-cloud performance and cost reduction ($r=0.45$), which means that, in the context of CDNs depending on multi-cloud configurations an organization would be better able to optimize its costs especially if they are capable of spreading loads across several service providers platforms.

Insights from the Statistical Analysis:

The negative coefficients in the regression suggest that DevOps practices were correlated with cost reduction and multi-cloud performance, -0.42 and -0.38 , respectively (Table 2). These results are encouraging the notion that the maturity of DevOps highly contributes to efficient cloud management. The benefit of DevOps in terms of cost reduction (p-value 0.001) is an indication that process automation and optimized workflows bring the costs down. Moreover, the

moderate correlation of DevOps with scalability (coefficient = 0.29) highlights specifically an increasing capacity's ability to scale cloud resources on-the-fly within an organization and is especially important for multi-cloud organizations.

The ANOVA test results provided additional support to these observations with readings of $p = 0.003$ and F-statistic of 5.32 for cloud provider and $p = 0.005$, F-statistic = 4.81 for industry indicating existence of statistically significant differences in multi-cloud performance across cloud providers and industries, respectively. These findings illustrates how the selection of cloud service provider and sector can significantly influence the success of multi-cloud initiatives. For example, the users of AMI gave top scores to performance since it has a well-developed ecosystem and an abundant amount of tools on DevOps and managing cloud infrastructure. The inter-sector variability, with the IT services sector performing better contrasted to other sectors, points out that sectoral characteristics could affect the proficiency of multi-cloud adoption (e.g., cloud expertise/ maturity).

Interpretation and Discussion with respect to the research Questions and Objectives:

The survey findings are very consistent with the research questions and objectives. RQ1 Aim: To study the effects of DevOps adoption on performance of cloud deployment. As the stats above show, companies that are more DevOps focused fare better in multi-cloud than those who aren't showing that DevOps clearly is a key enabler of 'cloud-optimisation'. RQ2: How do multi-clouds affect cost and scalability? The report also discovered that costs can go down — and performance up, further underlining advantages gained from using more than one vendor's cloud offering. Such conclusions provide the guidance for SMEs to drive them towards best cloud infrastructures and DevOps adaption.

4.3.2 Addressing Research Gaps

The results of this research provide valuable contribution to the DevOps adoption and multi-cloud strategies in SMEs, for which literature identifies many research gaps. Although current research has thoroughly investigated applications of DevOps in large organizations (Chen et al., 2024), a knowledge gap exists with respect to its direct implementation in SMEs. This research fills the gap by illustrating how SMES can benefit from multi-cloud performance, economics and scalability by adopting DevOps practices.

One of the main limitations in the current literature is that there is not enough consideration about how DevOps should be adopted in multi-cloud environment context inside SMEs. While researches have examined the advantages of DevOps in enhancing software delivery and operational effectiveness (Pochu et al., 2014), the use of this technology on various cloud environments for organizations with constrained resources has not been extensively studied. Addressing this gap, our work demonstrates that DevOps practices, such as (CI and [6]) directly impact better cloud quality performance and cost minimization for multi-cloud.

Similarly, it is also important to fill the gap by examining how multi-cloud adoption affects cost-efficiency in SMEs. Though, it has been reported that multi-cloud strategies provide more flexibility and robustness (Merseedi and Zeebaree, 2024) only few researchers have quantified the cost advantages that SMEs may enjoy using multi-cloud infrastructure. The study discovers a moderate relationship between multi-cloud performance and cost savings, which shows that those SMEs benefiting from the deployment of multi-cloud tactics can realize efficient resource allocation across multiple cloud platforms.

Moreover, this research helps understanding the issues faced by SMEs in multi-cloud adoption and deploying DevOps practices including skills shortage and integration complexity. As noted by Granlund et al. (2021) Toward SMEs in particular, might not have sufficient technical strength to effectively build complex multi-cloud systems. This research supports these conclusions, shedding light on the fact that skills gap and integration are still some of the primary challenges to SMEs multi-cloud adoption.

In conclusion, this research contributes to bridging literature gaps by investigating how DevOps and multi-cloud adoption intersect in SMEs, demonstrating a combined benefit through empirical findings. It does an excellent job of conveying to readers how small and medium enterprises can use DevOps best practices to get the most out of multi-cloud, control costs, and generally just improve their quality of service.

4.3.3 Literature Synthesis

4.3.3.1 Alignment with Existing Literature

The results of this study are rather consistent with the available literature on DevOps adoption and the multi-cloud strategy in the framework of SMEs. The same trend has been noted by other researchers, such as Pochu et al. (2024) who emphasize that DevOps practices allow improving

the speed of software delivery and operational efficiency, and the positive relationship between the practices of the DevOps and the multi-cloud performance that can be seen in the present study will corroborate the same. The fact that DevOps and cloud performance have a correlation of 0.68 confirms that organizations that have adopted DevOps practices will realize tangible changes in the way they conduct their cloud operations such as speedy delivery cycles and enhanced reliability.

Also, the correlation between multi-cloud performance and cost reduction is in line with the results by Merseedi and Zeebaree (2024) who indicated that multi-cloud strategies improve the optimization of costs, as it avoids vendor lock-in and improves the allocation of resources. The results of our study, in which a medium positive correlation (0.45) between performance and cost reduction was observed, resonate with these advantages of the multi-cloud adoption, revealing that the SMEs are able to utilize the advantages of multiple cloud providers to achieve higher performance and become cost-effective. These results also support the assertion of the literature about the potential of multi-cloud environment to potentially save costs.

The research is consistent with other studies that have done similar work, including one by Chen et al. (2020) proposed that DevOps practices could promote collaboration and efficiency in the team level. Our regression model indicated that DevOps significantly exerted a positive effects toward cost savings (coefficient = 0.42), which also supported by the previous literature in the terms of reduced cost resulted from automated routine and cooperation among organization in DevOps mode.

4.3.3.2 Contradictions with Previous Studies

While the study validates much of what has been found in previous literature, there are results that are different than what is currently available. An example of these is the effect that DevOps has on scalability. Other past studies, including the one by Waseem et al. (2025), implied that DevOps guidelines can be highly scaled, which is due to the automation of the infrastructure and control over resources. However, this study is also exposing a weak level of satisfaction (0.29) at the intersection of DevOps and scalability. It was significant statistically but had a smaller effect than anticipated. Some of this discrepancy may be attributable to differences in DevOps maturity among the organizations interviewed, or it could be since scaling a multi-cloud environment is hard perhaps too hard for DevOps alone.

Another distinction from previous research relates to the industry-specific performance of multi-cloud strategies. While Granlund et al. (2021) who found no significant differences at least between manufacturing and IT services which contrasts findings in this research that there is a difference at least between manufacturing industry and IT services with the former having lower performance than the latter in multi-cloud. The findings highlight a maturity gap in the adoption of cloud among manufacturing and other industries, where multi-cloud cannot gain traction effectively potentially pointing toward different degrees of cloud/DevOps competency within each industry.

4.3.3.3 Contributions to the Literature

This study makes several contributions to the literature for SMEs, and since it provides new perspectives on the combined impact of DevOps (R Tech) adoptions with multi-cloud performance. Firstly, it is one of the first empirical studies to demonstrate that DevOps enhances performance and cost efficiency in small-medium enterprise multi-cloud systems (SME) systems which have received less attention than large enterprises.

Second, the paper contributes an understanding of the interplay among multi-cloud cost and performance. While previous work has mostly focused on the cost advantage of the corresponding multi-cloud strategy for large organization (as opposed to the research approach taken by other articles with regard to savings in costs for enterprise level multi cloud benefit, we have found that small-medium enterprises can also obtain significant saving of their cloud usage bill if they adopt and use multi clouds from a performance angle and resource utilization perspective.

Finally, the research paper adds to the subtle comprehension of the impact of industry-specific elements on the effectiveness of the multi-cloud strategies. The study offers information that cloud maturity and technical skills are major determinants of the effectiveness of this approach in the adoption of DevOps and multi-cloud by identifying IT services as the most effective area of implementing these strategies. This observation prepares the ground to conduct future studies on industry-specific impediments and facilitators of multi-cloud usage and DevOps application among SMEs.

4.3.4 Addressing Research Objectives (ROs)

The research aims were addressed through a thorough survey analysis, statistical confirmation, and synthesis with previous literature. Organizational benefits toward achieving the first objective in investigating DevOps practices on multi-cloud performance was centering the relationship between adopting of DevOps and reported improvements. This positive association (correlation coefficient is 0.68) supported DevOps as a significant multi-cloud performance enabler.

The objective 2, which aims at knowledge of the impact of multi-cloud strategies on cost reduction, was answered by the determination of a moderate level of relationships between performance and cost savings (0.45). The regression analysis also identified the direct effect of DevOps in reducing cost, which establishes the influence of this initiative in optimizing cloud infrastructure.

Finally, Objective 3 to explore the contribution of industry-specific factors was attained through ANOVA, which showed that the performance differed significantly across industries, especially in IT services and manufacturing industries.

4.3.5 Addressing Research Questions (RQs)

Research Question 1 was to determine the effect of DevOps implementation on the multi-cloud performance. The results demonstrate that there is a positive correlation (0.68), so the more mature DevOps practice organizations are, the more likely they are to report low multi-cloud performance, which proves the importance of DevOps in improving the cloud operations.

Research Question 2 was based on the research of the impact of multi-cloud strategies on cost minimization. The research revealed that multi-cloud performance and cost-saving show a moderate positive correlation (0.45) implying that multi-cloud environments assist SMEs to maximize costs by allocating resources more efficiently.

The 3rd Research Question examined the influence of industry-specific factors on cloud performance. The outcomes of the ANOVA revealed that the difference was significant, with the IT services being more successful in the reported performance than manufacturing, which indicates that the success of the multi-cloud adoption is dependent on the industry maturity and cloud expertise.

4.3.6 Implications for SMEs

The results provide a few practical guidelines for SMEs that wish to improve their multi-cloud and DevOps. In the first place, SMEs must have DevOps on their top list of priorities as it plays a vital role in accelerating multi-cloud performance and reducing costs. Automate workflows and make collaboration easy with SME, so you can run efficiently and reduce operational costs.

Two, SMEs should look at a multi-cloud approach to prevent vendor lock-in and become more flexible. Leveraging the performance in a multi-cloud environment results in resource saving and cost reduction.

Lastly, it is important to invest in team skills for DevOps and multi-cloud management because lack of skills was cited as one of the most significant roadblocks. Higher performing IT services organizations should act as a guide for other industries, also specifically manufacturing to advance their cloud and DevOps maturity.

CONCLUDING REMARKS

This study addressed the need to understand GH businesses when leveraging DevOps Practices and Multi-Cloud Strategy affect on Cloud runtime, cost savings, and elasticity. This paper has been helpful for understanding how SMEs use these technologies and practices in terms of migrating to the cloud infrastructures, upgrading their cloud resources or improving their operational efficiency and force productivity by outputting services.

Summary of Key Findings

The survey results have shown that DevOps practices has significant impact in multi-cloud performance optimization and cost reduction on SMEs. It was found a strong positive relationship between DevOps adoption with better cloud performance and higher level of DevOps maturity by the organizations correlated also to faster deployment times, better system stability and lower operational costs. And on top of it — multi-cloud is about to make it easier for companies to start reducing costs by enabling smarter resource allocation and stopping the stranglehold of a vendor-monopoly.

Additionally, there were sectoral trends that emerged in the study. IT services companies also deploy the best performance in the cloud, which could be because these firms have more mature DevOps and cloud skills. Multi-cloud configurations, on the other hand, proved more challenging for manufacturing firms to seamlessly adopt. The results showed that both DevOps and multi-cloud uptake and success is driven by industry maturity and technical capability.

Our regression analysis has shown that DevOps has positive impact on cost efficiency and cloud performance, while there is a significant association between DevOps maturity levels and cost saving. ANOVA has sustained further evidence for performance between industries issue: IT Services / Software development lead the pack in adoption of cloud/DevOps-related technologies.

Recommendations

The findings from the report also reveal some useful recommendation for SMEs aiming to use multi cloud and DevOps effectively:

- **Critical Success Factor:** Deploy DevOps Practices and Exploit their Advances SMEs have to dedicate time and resource into deploying or escalating best practice demands within the DevOps, as the long-term effects on cloud performance will be substantial. By identifying automation flows, establishing run-book processes and encouraging teamwork across devops teams SME's can enable efficiencies -operational efficiencies which stomp out those manual errors.
- **Embrace a Multi-Cloud Strategy:** The results prove that multi-cloud is essential to minimizing risk and dependence on any single vendor. Leveraging several cloud providers makes the SME more flexible, and more resilient in a situation to better control costs. SMEs must match their cloud provider to the specific requirements of their workloads and ensure that all systems and platforms can be integrated for smooth business processes.
- **Skills and Training:** A big impediment to effectively executing DevOps and multi-cloud is the lack of available talent. Training to Succeed SMEs must train and upskill their talent on cloud management and DevOps principles. It is critical for solving integration-issues and achieving friction-less execution of mature cloud tactics.
- **Industry-Specific Problems:** SMEs in various industries face certain issues of embracing DevOps and multicloud. For example, IT services companies may already have deeper knowledge and expertise around advanced cloud strategies, whereas manufacturing might require a bespoke approach with more hand-holding. There must be industry-specific model templates for the less mature verticals to follow multi-cloud and/or DevOps best practice.

In order to eliminate operational inefficiencies, SMEs need to aim at automating operations with such a tool as Jenkins to CI/CD, Kubernetes to manage containers, and Terraform to provide infrastructure. Also, the incorporation of DevSecOps, such as the use of Snyk and Aqua Security, will make security a self-maintaining and automated process and not something that comes in the end.

Future Research Directions

Even though there are some valuable conclusions in this research, it seems still not exhausted several other relevant elements to be investigated as well:

- **Longitudinal DevOps and Multi-Cloud Adoption Effect Studies:** Future work could feature long-term follow-up studies to examine the long-lasting effects of adopting DevOps

and multi-cloud over multiple years towards determining the extent of ongoing advantages in terms of cost-effectiveness, scalability, and performance. In order to gain deeper understanding of long-term development and success, it would be interesting to follow firms over time.

- **Sector-Specific Study:** The study implies that sector-specific barriers and enablers have more to be learned in adopting DevOps/multi-cloud. An example of this is the supply requirements by the suppliers of goods might not be similar to those of IT services or healthcare providers. And, speaking of which industries follow and gain advantages through such practices, we may gain some valuable lessons concerning personalization.
- **Adopting emerging technologies:** Future work might also explore the intersection of emerging technologies like AI, machine learning or serverless computing with the DevOps and multi-cloud strategies to improve the best cloud performance and operational efficiency in SMEs. Such technologies will be the basis in the automation and optimization methods to manage cloud services in large scale.
- **Comparative Study on Global vs Regional Strategies:** Comparative analysis of global vs. regional trends in adoption of DevOps/multi-cloud strategy may help understanding how regulatory compliance and maturity of market/cloud infrastructure availability can influence the best practices for adopting such strategies. This will enlighten SMEs on regional matters and their cloud strategy choices.

Final Thoughts

To conclude, this thesis has given some insights on how DevOps is applied and how multi-cloud strategy can contribute to better cost effectiveness, performance agility in SMEs. The results showcase the value of these practices in enhancing efficiency and boosting growth. Although the SMEs face challenges such as skills and integration, in this paper we provide practical recommendations for overcoming these constraints and making the most of what cloud may offer.

As digital transformation becomes even more entrenched in people's thinking, it's obvious that it will be of importance that small businesses continue to focus on rolling-out DevOps and the adoption of multi-cloud strategies for this who know small projects don't fit into these cloud services.

The SMEs can implement the following tools and frameworks to improve their DevOps and multi-cloud strategies:

- **In the case of DevOps Automation:** SMEs may use continuous integration and continuous deployment (CI/CD) pipelines with the help of such tools as Jenkins. This will aid in automation of the software deployment process and minimized errors and enhanced deployment speed.
- **Cloud Resource Management:** Cloud providers, such as Terraform and Ansible may be used to systematically provide and set up cloud resources to ensure uniformity and reduction in the manual load of operating in a multi-cloud setting.
- **To perform Monitoring and Performance Optimization:** Prometheus and Grafana could be incorporated into the DevOps pipeline to monitor and analyse the performance of applications in more than one cloud environment in real-time to enable SMEs to determine the bottlenecks and optimize resource usage.
- **Multi-Cloud Cost Management:** Cloud Health by VMware or Spot.io would be useful to SMEs to monitor and optimize cloud costs, as it will give them an understanding of cloud spending with different providers and make better decisions regarding the allocation of resources.
- **To comply with the Security:** To implement Security into the DevOps cycle, it is possible to utilize DevOps solutions like Aqua Security or Sync that will help both identify vulnerabilities at the earliest stages of the development life cycle and deal with them within the contexts of multi-clouds.

Through these tools, the SMEs would be able to promote collaboration, increase efficiency and handle the complexities of the multi-cloud environment better. These tools can also help SMEs embrace a more integrated and simplified perspective of DevOps so that they can effectively scale at an affordable cost and performance.

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APPENDIX

Appendix A - Survey Questionnaire

Section 1: General Information

1. What is your role in the organization?

☐ IT Manager

☐ Cloud Architect

☐ DevOps Practitioner

☐ Other (Please specify)

2. How many employees are in your organization?

☐ 1-50

☐ 51-200

☐ 201-500

☐ 501+

3. Which cloud providers are you currently using? (Select all that apply)

☐ AWS

☐ Microsoft Azure

☐ Google Cloud

☐ IBM Cloud

☐ Other (Please specify)

4. How long has your organization been using a multi-cloud environment?

☐ Less than 1 year

☐ 1-3 years

☐ 4-5 years

☐ More than 5 years

Section 2: DevOps Adoption and Practices

5. Has your organization adopted DevOps practices?

☐ Yes

☐ No

6. Which DevOps practices are currently implemented in your organization? (Select all that apply)

- ☐ Continuous Integration (CI)
- ☐ Continuous Deployment (CD)
- ☐ Automated Testing
- ☐ Infrastructure as Code (IaC)
- ☐ Monitoring and Logging
- ☐ DevSecOps (Security integration in DevOps)

7. How would you rate the overall effectiveness of DevOps in your organization?

- ☐ Very Effective
- ☐ Effective
- ☐ Neutral
- ☐ Ineffective
- ☐ Very Ineffective

8. What challenges have you faced in implementing DevOps practices?

- ☐ Lack of skilled staff
- ☐ Integration with existing systems
- ☐ High initial setup costs
- ☐ Security concerns
- ☐ Other (Please specify)

9. To what extent has DevOps improved the speed of software delivery in your organization?

- ☐ Greatly Improved
- ☐ Moderately Improved
- ☐ No Change
- ☐ Decreased Delivery Speed
- ☐ Not Sure

10. How often do you update your cloud infrastructure through DevOps processes?

- ☐ Daily
- ☐ Weekly
- ☐ Monthly
- ☐ Quarterly
- ☐ Less frequently

Section 3: Multi-Cloud Environment

11. How would you rate the overall performance of your multi-cloud environment?

- ☐ Excellent
- ☐ Good
- ☐ Fair
- ☐ Poor
- ☐ Very Poor

12. How important is multi-cloud adoption for your organization?

- ☐ Very Important
- ☐ Important
- ☐ Neutral
- ☐ Not Important
- ☐ Not Sure

13. To what extent has using a multi-cloud environment improved your organization's operational flexibility?

- ☐ Greatly Improved
- ☐ Moderately Improved
- ☐ No Change
- ☐ Decreased Flexibility
- ☐ Not Sure

14. What benefits have you experienced by adopting a multi-cloud environment? (Select all that apply)

- ☐ Vendor lock-in avoidance
- ☐ Cost optimization
- ☐ Better performance
- ☐ Scalability
- ☐ Risk management and redundancy
- ☐ Increased flexibility
- ☐ Other (Please specify)

15. What challenges have you faced in managing a multi-cloud environment?

- ☐ Complexity of integration
- ☐ Data security issues

- ☐ Increased costs
- ☐ Lack of in-house expertise
- ☐ Performance issues
- ☐ Other (Please specify)

16. How do you manage the orchestration and integration of different cloud providers?

- ☐ Manual configuration
- ☐ Automated tools
- ☐ Hybrid approach
- ☐ Not sure

17. To what extent do you experience downtime or performance degradation in your multi-cloud setup?

- ☐ Frequently
- ☐ Occasionally
- ☐ Rarely
- ☐ Never

18. How well do your cloud service providers integrate with each other?

- ☐ Seamlessly
- ☐ With some difficulties
- ☐ Poorly
- ☐ Not Sure

Section 4: Cost Efficiency and Optimization

19. Has implementing a multi-cloud environment led to cost savings for your organization?

- ☐ Yes, significant savings
- ☐ Yes, moderate savings
- ☐ No change
- ☐ Increased costs
- ☐ Not sure

20. How does your organization optimize costs across multiple cloud providers?

- ☐ Manually adjusting resources
- ☐ Using automated tools
- ☐ No specific cost optimization strategies

☐ Not sure

21. How do you assess the cost-effectiveness of your cloud providers?

☐ Regular cost audits

☐ Use of cost management tools

☐ Based on performance and scalability

☐ No specific assessment strategy

22. To what extent has DevOps helped reduce cloud infrastructure management costs?

☐ Greatly Reduced

☐ Moderately Reduced

☐ No Change

☐ Increased Costs

☐ Not Sure

23. How would you rate the ability of your organization to scale cloud resources as per demand?

☐ Very Scalable

☐ Moderately Scalable

☐ Not Scalable

☐ Not Sure

24. Do you monitor and adjust your cloud infrastructure in real-time?

☐ Yes, continuously

☐ Yes, occasionally

☐ No

☐ Not sure

Section 5: Security

25. How would you rate the security of your multi-cloud environment?

☐ Very Secure

☐ Secure

☐ Neutral

☐ Insecure

☐ Very Insecure

26. How often do you review security measures across your multi-cloud environment?

☐ Daily

- ☐ Weekly
- ☐ Monthly
- ☐ Quarterly
- ☐ Less frequently

27. Has DevOps integration improved your ability to address security concerns in the multi-cloud environment?

- ☐ Greatly Improved
- ☐ Moderately Improved
- ☐ No Change
- ☐ Made Security Worse
- ☐ Not Sure

28. What are the biggest security risks you face in your multi-cloud environment?

- ☐ Data breaches
- ☐ Inadequate encryption
- ☐ Lack of compliance with regulations
- ☐ Integration vulnerabilities
- ☐ Other (Please specify)

29. Do you use any DevSecOps practices (security integrated within DevOps) in your multi-cloud environment?

- ☐ Yes
- ☐ No
- ☐ Not Sure

30. How confident are you that your multi-cloud environment meets industry security standards?

- ☐ Very Confident
- ☐ Confident
- ☐ Neutral
- ☐ Not Confident
- ☐ Not Sure

this is primary research and a dataset for a thesis. Here is the GitHub link:
[https://github.com/Lokesh0423/survey-data-
upload/blob/main/survey_responses_with_questions.csv](https://github.com/Lokesh0423/survey-data-upload/blob/main/survey_responses_with_questions.csv)